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# Paleomagnetism and Geology of Eocene Volcanic Rocks of Southwest Washington, Implications for Mechanisms of Tectonic Rotation

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Paleomagnetic and geologic investigations in Eocene volcanic rocks of the southwest Washington Coast Range demonstrate a close relationship between tectonic rotations and the local structural geology. The allochthonous middle Eocene submarine basalt basement of the Crescent Formation consists of several fault-bounded structural domains up to 30 km across that are characterized by different amounts of clockwise rotation (20° to perhaps as much as 65°) when compared to the Eocene reference pole for North America. Structural analysis shows the differential rotations postdate middle Eocene folding of the Crescent Formation against the continental margin and predate the unconformably overlying upper Eocene Goble Volcanics, which are rotated about 23° and do not show the same domains of rotation as the underlying Crescent Formation. Post-Goble rotations may be accommodated by a fault pattern very similar to that expected for areas caught in a simple dextral shear couple along transcurrent faults. Major north-northwest trending faults with several kilometers of dextral displacement form the boundaries of cross-faulted shear domains in which the clockwise rotation of elongate crustal slices is accommodated by west-northwest trending sinistral *R'* Riedel shears. Thirty-five to 100% of the observed post–late Eocene rotations could have occurred by this shear rotation mechanism. Other paleomagnetic study areas in the Coast Range and western Cascades have a similar fault geometry and may also have undergone significant shear rotations. Long-term northward oblique subduction of the Farallon plate beneath the Coast Range throughout most of the Tertiary could have been the driving force for the shear rotations and could explain the rapid eastward decrease in rotation away from the continental margin. Shear rotations could eliminate many of the structural and stratigraphic difficulties associated with models involving rotation of large, rigid plates.

## 1. INTRODUCTION

Paleomagnetic and geologic studies in the Western Cordillera show that large-scale rotation and translation of crustal fragments have occurred along the Pacific margin of North America since at least Jurassic time [*Beck*, 1980; *Coney et al.*, 1980]. These studies have generated new interest in the tectonic significance and kinematic history of these enigmatic mobile fragments, and many of these "suspect" terranes have been the subject of intensive paleomagnetic and geologic investigations. The early Tertiary Oregon-Washington Coast Range may be one of the youngest allochthonous terranes making up the accretionary collage of western North America (Figure 1). Thick submarine and subaerial basalts that form the Coast Range basement have been interpreted as oceanic islands, seamounts, and ocean floor that have been accreted to the continent sometime during the Eocene [*Snavely and MacLeod*, 1974; *Dickinson*, 1976; *Simpson and Cox*, 1977].

Considerable paleomagnetic evidence now documents rotation of the Oregon Coast Range [*Simpson and Cox*, 1977; *Beck and Plumley*, 1980; *Magill et al.*, 1981]. Paleocene to middle Eocene submarine basalts of the Oregon Coast Range basement show 70° of clockwise rotation, while successively younger rocks show lesser amounts of rotation. *Beck* [1976, 1980] has suggested that anomalous eastward magnetization directions characteristic of the Cordillera may be the result of clockwise rotations of crustal blocks in a broad zone of right-lateral shear generated by the relative northward motion of oceanic plates to the west. In southern California, *Luyendyk et al.* [1980] have shown a correlation between areas of clockwise rotation and fault geometry which suggests that simple shear rotation of large fault-bounded slivers has occurred. In western Oregon and Washington, *Sheriff and Bentley* [1980] have proposed that Neogene clockwise rotations are the result of distributed shear acting on closely spaced right-lateral faults of small displacement. Alternatively, *Simpson and Cox* [1977] and *Magill et al.* [1981, 1982] have proposed several rigid plate models for the clockwise rotation of the Oregon-Washington Coast Range in which an elongate Coast Range microplate rotates during collision with North America and/or during asymmetric back arc spreading following accretion.

The picture that is evolving from recent paleomagnetic studies in the Washington Coast Range suggests that Washington is rotated much less than Oregon and may be part of a separate oceanic fragment [*Beck and Burr*, 1979; *Wells and Coe*, 1979, 1980; *Beck and Engebretson*, 1982; *Globerman et al.*, 1982]. The Willapa Hills in the southwestern Washington Coast Range occupy a "keystone" position in the center of the Oregon-Washington Coast Range near the probable boundary between the two terranes and are thus a prime target for paleomagnetic and geologic investigations.

We have studied the Eocene volcanic rocks of the Willapa Hills using a combination of paleomagnetic sampling, geologic mapping, and structural analysis. We collected samples from several basement uplifts of the middle Eocene Crescent Formation and also from the unconformably overlying upper Eocene Goble Volcanics. Our investigation addresses two basic questions:

1. How large are the rotated blocks? Are they microplates

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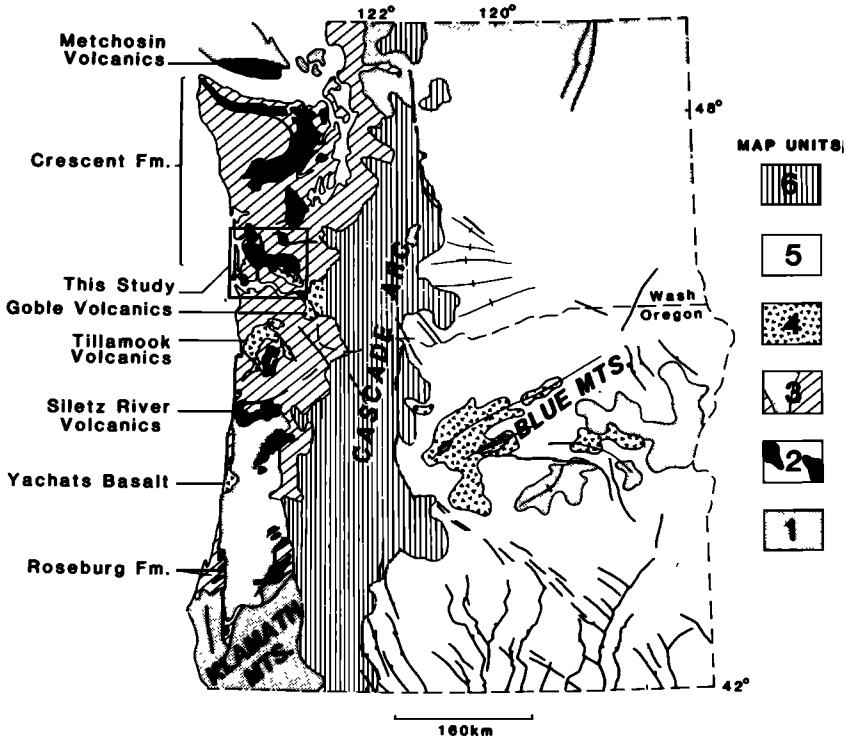



Fig. 1. Geologic and tectonic map of the Pacific Northwest after *King and Beikman* [1974]: 1, pre-Tertiary rocks; 2, Paleocene and Eocene oceanic basalt of Coast Range basement; 3, mostly marine Tertiary sedimentary rocks, as mapped includes Quaternary deposits in Puget Sound region, Eocene Tyee Formation stippled; 4, Eocene and early Oligocene volcanic rocks, includes basaltic rocks of the Coast Range and Clarno Formation in central Oregon; 5, mostly Oligocene and Miocene volcanic rocks of the Columbia Plateau and Oregon Basin and Range; 6, Tertiary and Quaternary volcanic rocks of the Cascade arc.

the size of the Oregon Coast Range or western Washington and Oregon or are they small blocks a few kilometers across? Solving this problem would have significant impact on paleogeographic reconstructions based on paleomagnetic data.

2. What is the nature of the tectonic boundary between rotated blocks? Are they strike-slip faults, thrust faults, zones of melange, or diffuse zones of plastic deformation? Defining a relationship between structure and paleomagnetic results would give us a better understanding of the mechanisms of rotation and their relationship to regional tectonic processes. In this paper, tectonic rotations can be related to specific geologic structures, and we examine the implications for microplate rotation models.

### *Geologic Sketch of Southwest Washington*

The major tectonic feature of southwest Washington is the large arcuate basement high that forms the backbone of the Willapa Hills (Figure 2). Exposed in the core of this complex uplift are middle Eocene submarine basalts of the Crescent Formation, the oldest rocks in southwest Washington. The Crescent Formation is composed primarily of submarine pillow basalt flows and thin siltstone interbeds [*Wolfe and McKee*, 1972; *Wells*, 1981]. *Duncan* [1982] reports an average K-Ar age for the basalts of about 49 m.y. In the upper part of the formation, pillow basalts are interbedded with abundant basalt breccia, basaltic sandstone, and conglomerate, some of which contain shallow water marine fossils. Subaerial basalt flows and interbedded basalt boulder conglomerate occur locally at the top of the section. In other places the upper part of the Crescent Formation is interbedded with and conformably overlain by middle Eocene deepwater siltstone, and the basalt and siltstone are intruded by an extensive sill complex of middle (?) Eocene gabbros [*Wells*, 1981]. A thick sequence of upper middle Eocene to Miocene marine siltstone and sandstone unconformably overlies the older rocks.

The total thickness of the Crescent Formation cannot be determined because the base is not exposed. We estimate it is at least 3 km thick in southwest Washington. Geochemically, the basalts are characterized by relatively high titanium and alkali contents, similar to the more differentiated caps of alkali lavas found on oceanic islands. For these reasons we interpret the Crescent Formation in southwest Washington to be part of an oceanic island province similar to that suggested for the partly correlative Siletz River Volcanics in the Oregon Coast Range [*Snavely et al.*, 1968] and for parts of the Crescent Formation in the Olympic Mountains and in the Black Hills of Washington [*Cady*, 1975; *Globerman and Babcock*, 1980].

### *Structure*

The ridge and basin structure of southwest Washington trends northwest across the Coast Range axis, nearly at right angles to the northeast trend of basement uplifts in the Oregon Coast Range (Figure 1). In Washington, the basement uplifts form asymmetric anticlinoria with vertical or overturned western limbs possibly bounded by landward dipping thrust or reverse faults [*Pease and Hoover*, 1957; *Snavely et al.*, 1958; *Gower and Pease*, 1965; *Wells*, 1979, 1981; *Snavely and Wagner*, 1982]. Bouguer gravity and aeromagnetic gradients [*Bonini et al.*, 1974; *Zietz et al.*, 1971] coincide with the fault-bounded margins of the basalt basement uplifts and define beneath the sedimentary veneer the probable outlines of discrete tectonic "blocks" in the Washington Coast Range.

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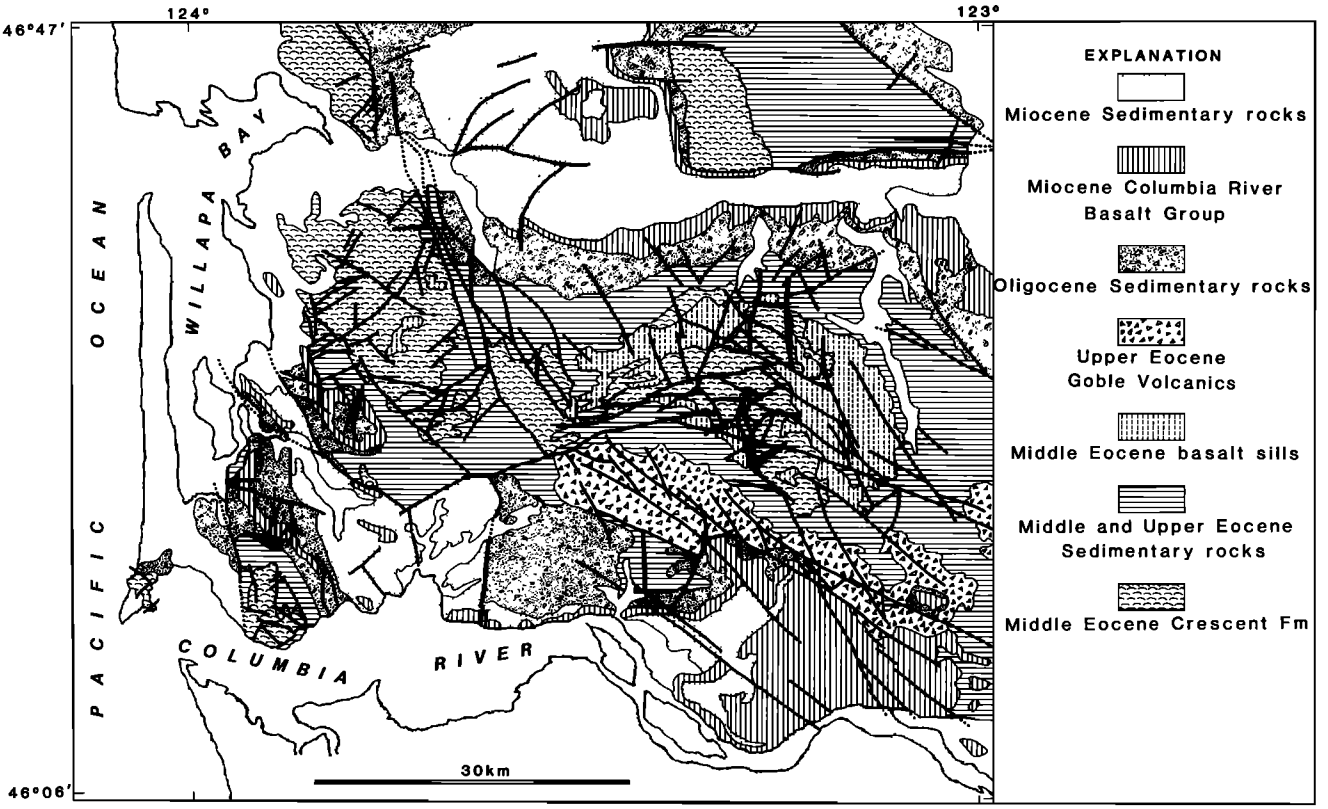



Fig. 2. Geologic map of southwest Washington from *Pease and Hoover* [1957], *Wagner* [1967*a*, *b*], *Wolfe and McKee* [1968], and *Wells* [1979, 1981]; solid lines are faults.

The ridge and basin structure of Washington contrasts with the broad Oregon Coast Range uplift and its relatively continuous gravity high [see *Magill et al.*, 1981].

The regional angular unconformities in the marine sedimentary sequence of the Coast Range define two major periods of deformation during the Eocene and middle Miocene [*Snavely and Wagner*, 1963; *Snavely et al.*, 1980]. In southwest Washington, deformation of the Crescent basalts began during the middle Eocene with the formation of northwest trending open folds that become nearly isoclinal along the western margins of the basement uplifts (Figure 3). Superimposed locally on the northwest set are northeast trending open folds also of Eocene age. This early folding occurred during the middle Eocene, as judged from the onlap of the late middle and upper Eocene Cowlitz Formation and Goble Volcanics [*Wells*, 1981] and the angular discordance between the trends of the basement folds and those in younger sedimentary rocks. Subsequent late middle Miocene and younger (?) deformation in the basinal sediments produced west-northwest open folds that become more northwesterly near the coast, where they are nearly coaxial with the folds in the Crescent basement. This later deformation was accentuated near the margins of block uplifts where the Miocene sedimentary rocks are in places steeply dipping to overturned.

The fault pattern in southwest Washington is complex and subject to interpretation because of poor exposures, but four major fault sets can be recognized (Figure 3):

1. North 20°–40°W faults: occur along oversteepened limbs of large uplifts and are generally vertical to east dipping with significant vertical offset (about 1 km?); dextral slip component is indicated by slickensided and grooved fault surfaces (Figure 4*a*); they are commonly associated with minor conjugate left-lateral faults trending N40° to 60°E.

2. North 60°–70°W faults: mostly south of the Naselle-Grays River fault system; occur as vertical faults with several tens or hundreds of meters of normal stratigraphic separation, although grooved fault surfaces suggest horizontal latest motions (Figure 4*b*).

3. North 45°–80°E faults: form left-stepping nearly vertical faults of the Grays River fault zone along the southern margin of the Willapa Hills uplift, and the high-angle Doty fault forms the southern boundary of the Doty uplift; also includes closely spaced faults north of the Naselle-Grays River fault zone. Stratigraphic separations on the Grays River faults indicate down to the south, but again horizontal grooves predominate on fault surfaces, and the faults show about 2-km sinistral offset of earlier formed structures.

4. North 10°–30°E faults: occur as short segments between longer northwest faults; have grooves, slickensides, and stratigraphic separations consistent with normal faulting.

The nearly vertical fault surfaces and the abundant indicators of horizontal motion suggest that strike-slip faulting played an important part in the kinematic history of southwest Washington. Although conjugate strike-slip faults can be demonstrated on an outcrop scale, the large-scale fault pattern is more complex. The lack of a simple fault geometry and the greater complexity of faulting in older rocks show that several deformations have occurred, most notably during the late middle Eocene and middle Miocene. Some faults which cut through the entire stratigraphic section have had a long history, with the greatest offset and deformation associated with the oldest rocks. The reactivation of older structures may ex-

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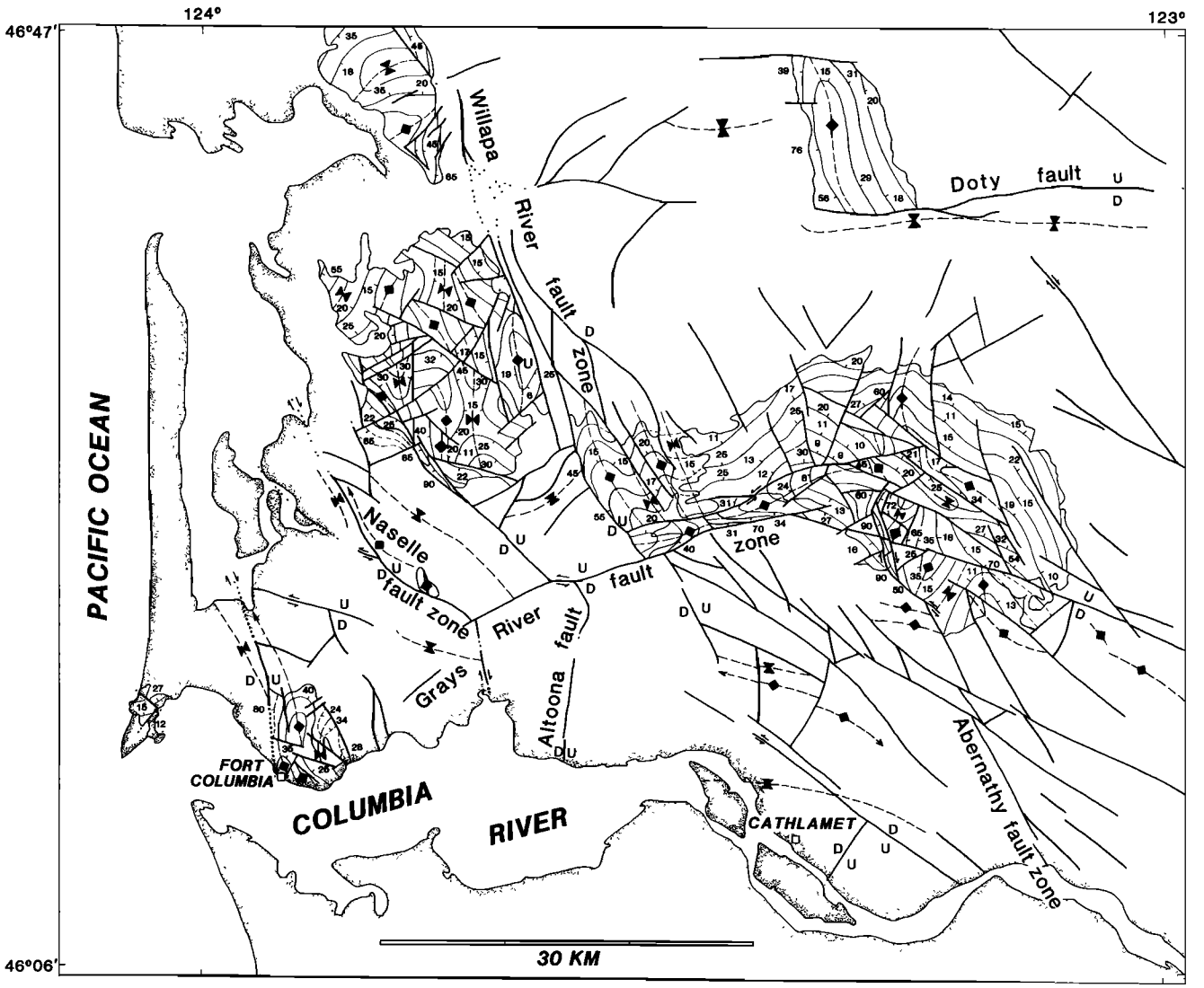



Fig. 3. Structure map of southwest Washington from *Pease and Hoover* [1957], *Snavely et al.* [1958], *Gower and Pease* [1965], *Wagner* [1967*a*, *b*], *Wolfe and McKee* [1968], and *Wells* [1979, 1981, unpublished mapping]. Heavy lines are faults; stratigraphic separations shown by U, D symbols; arrows show sense of lateral displacement; dashed lines are fold axes with diamond symbol for anticlines and bowtie for synclines; fine lines are schematic bedding plane traces as seen on a horizontal section through the Crescent Formation; tick marks and numbers show direction and dip of bedding in degrees.

plain the geographic provinciality of different fault sets and their lack of conformity to the expected regional north-south maximum horizontal compressive stress indicated for the Pacific Northwest during the Neogene [*Zoback et al.*, 1981]. We might expect the strain history of the Coast Range to be complex because of deformation during its emplacement and subsequent changes in relative plate motions. In a later section we will speculate on the kinematic significance of some of these faults in light of the paleomagnetic data.

## 2. Paleomagnetism of the Crescent Formation

### *Sampling and Laboratory Methods*

Our 76 site localities in the Crescent Formation were distributed throughout seven structural blocks generally separated from one another by large faults. Most of these blocks are part of the 100-km-long, arcuate Willapa Hills uplift (Figure 2). Geologic mapping at 1:62,500 scale [*Wagner*, 1967*a*, *b*; *Wolfe and McKee*, 1968; *Wells*, 1979, 1981] provided regional structural and stratigraphic control for the sites, and abundant siltstone interbeds in the pillow basalts provided good estimates of the local bedding attitudes at each site. With the exception of one streambed outcrop, our sites were entirely located in numerous quarries and roadcuts resulting from logging operations.

We sampled three rock types in the Crescent Formation: (1) pillow basalt flows and pillowed carapaces of thick flows; (2) columnar jointed interiors of thick submarine flows, and (3) columnar jointed to massive basalt and gabbro sills intruding the pillow basalts. Seven cores per site were collected using the methods of *Doell and Cox* [1965]. The cores were oriented with a magnetic compass, and readings were corrected for local anomalies by sighting on distant objects of known azimuth. We measured the remanent magnetizations of our samples and demagnetized them in alternating fields in the Stanford University cryogenic magnetometer. At least two cores per site were stepwise cleaned in alternating fields (AF) up to 100 milliteslas (mT) and vector demagnetization and stereonet diagrams were plotted concurrently. The other five cores per site were then stepwise cleaned in optimum fields determined

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Fig. 4*a*



Fig. 4*b*

Fig. 4. (*a*) Photograph of exposed N20°W trending vertical fault plane in basalt of the Crescent Formation, southwest Washington. Grooves and slickensides are nearly horizontal and suggest right-lateral displacement. (*b*) Photograph of N70°W vertical fault plane in Crescent Formation basalt. Grooves and slickensides plunge 7° to the northwest, but sense of lateral movement cannot be determined.

from the demagnetization plots, usually 10, 20, and 30 mT. For comparison with the results of the AF cleaning, two samples from each of 10 sites were stepwise thermally demagnetized up to 600°C, and the remaining cores from two sites were cleaned at 450°C.

*Results*

Results from three of the 76 sites were discarded because the magnetic directions failed to converge to a stable direction during stepwise demagnetization. Mean directions for the remaining 73 sites are listed in Table 1. NRM intensities averaged about $5 \times 10^{-2}$ A/m, with a maximum range of $5 \times 10^{-1}$ to $5 \times 10^{-4}$ A/m. Pillow basalt flows with pillows less than 1 m in diameter on the average had weaker NRMs ($10^{-1}$ to $10^{-3}$ A/m). NRM directions were quite scattered, but the basalts responded well to alternating field demagnetization (Figures 5 and 6). The median destructive field (MDF) ranged from 5 to 30 mT for most samples, with pillow basalts having higher MDF's (>15 mT) than columnar jointed flow interiors or sills (5 to 20 mT). Minimum dispersion of sample directions within a site was usually achieved with fields of 20 to 30 mT (Table 1). Many of the samples contained large secondary components that were easily removed with 10- to 15-mT demagnetizing fields. Secondary components of normal polarity can be attributed to viscous remanence (VRM) acquired in the present field, but some samples showed consistent secondary components of reversed polarity that suggested a second partial thermal or chemical remagnetization following the initial acquisition of thermoremanent magnetization (TRM).

The results from thermal demagnetization of 30 samples agree very well with the AF demagnetization of samples from

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TABLE 1. Paleomagnetic Data From the Crescent Formation, Southwest Washington, Tectonically Corrected Site Means

| Site | Demagnetization Field, mT | Site Location | | Remanence Direction | | VGP Position | | Statistics | | | | | Strike and Dip | Rock Type |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Latitude | Longitude | Inclination | Declination | Latitude | Longitude | $N$ | $k$ | $\alpha_{95}$ | ASD | Polarity | | |
| | | | | | *Domain 1* | | | | | | | | | |
| 8W207 | 20 | 46.76 | 236.17 | 35.7 | 60.1 | 34.6 | 333.9 | 7 | 97 | 6.2 | 8.2 | N | 030/35 | cjf |
| 8W200* | 20 | 46.77 | 236.30 | −68.8 | 65.2 | −23.6 | 018.8 | 7 | 588 | 2.5 | 3.4 | R | 348/49 | sill |
| 8W192 | 30 | 46.73 | 236.18 | 43.1 | 67.5 | 33.1 | 323.6 | 7 | 177 | 4.5 | 6.1 | N | 030/18 | pb |
| 7W163 | 10 | 46.74 | 236.22 | −53.5 | 180.1 | −77.3 | 235.8 | 6 | 23 | 14.4 | 17.0 | R | 270/20 | sill? |
| 7W155 | 10 | 46.71 | 236.22 | −76.5 | 202.5 | −68.5 | 83.1 | 6 | 12 | 19.9 | 23.1 | R | 030/45 | pb |
| 7W148 | 30 | 46.69 | 236.24 | −55.9 | 195.9 | −74.4 | 181.4 | 7 | 148 | 5.0 | 6.7 | R | 025/65 | pb |
| | | | | | *Domain 2* | | | | | | | | | |
| 7W170 | 30 | 46.61 | 236.13 | −68.5 | 138.3 | −62.7 | 352.5 | 7 | 40 | 9.7 | 12.9 | R | 050/25 | cjf |
| 7W062 | 30 | 46.60 | 236.27 | −23.8 | 205.7 | −49.6 | 195.5 | 6 | 28 | 12.9 | 15.3 | R | 0/0 | pb |
| 7W068 | 30 | 46.58 | 236.27 | −45.1 | 162.8 | −65.9 | 276.5 | 7 | 349 | 3.2 | 4.3 | R | 180/10 | pb |
| 8W239 | 30 | 46.54 | 236.17 | 58.0 | 33.3 | 64.5 | 331.8 | 6 | 36 | 11.3 | 13.5 | N | 320/20 | pb |
| 7W055 | 30 | 46.53 | 236.27 | 61.0 | 354.7 | 84.1 | 98.4 | 7 | 142 | 5.1 | 6.8 | N | 080/15 | pb |
| 7W032 | 20 | 46.51 | 236.19 | −45.1 | 163.7 | −66.3 | 274.8 | 7 | 131 | 5.3 | 7.1 | R | 140/35 | pb |
| 7W177 | 20 | 46.51 | 236.30 | 46.9 | 20.1 | 65.8 | 8.7 | 7 | 80 | 6.8 | 9.1 | N | 0/0 | pb |
| 7W039 | 20 | 46.50 | 236.21 | 64.1 | 97.5 | 27.2 | 287.2 | 5 | 129 | 6.8 | 7.1 | N | 120/36 | pb |
| 7W046 | 20 | 46.49 | 236.25 | 67.5 | 346.6 | 80.3 | 174.7 | 6 | 33 | 11.9 | 14.2 | N | 153/45 | pb |
| 7W184 | 30 | 46.49 | 236.32 | 56.0 | 4.0 | 79.6 | 38.2 | 7 | 130 | 5.3 | 7.1 | N | 110/30 | cjf |
| 7W191 | 99 | 46.48 | 236.31 | 52.1 | 47.3 | 51.7 | 330.2 | 8 | 18 | 11.6 | 19.0 | N | 110/30 | pb |
| | | | | | *Domain 3* | | | | | | | | | |
| 9W072 | 20 | 46.49 | 236.46 | 40.7 | 17.8 | 62.7 | 18.7 | 7 | 221 | 4.1 | 5.5 | N | 228/31 | pb |
| 9W080 | 30 | 46.49 | 236.46 | 71.0 | 338.2 | 73.7 | 187.8 | 7 | 33 | 10.7 | 14.2 | N | 240/25 | pb |
| 7W076 | 30 | 46.48 | 236.47 | 78.3 | 23.2 | 65.8 | 258.0 | 6 | 104 | 6.6 | 8.2 | N | 320/04 | sill? |
| 9W087 | 30 | 46.47 | 236.51 | 58.1 | 19.8 | 73.6 | 347.3 | 7 | 63 | 7.7 | 10.2 | N | 290/13 | cjf |
| 7W091 | 30 | 46.47 | 236.47 | 61.2 | 17.8 | 76.6 | 338.3 | 5 | 33 | 13.5 | 14.1 | N | 350/24 | pb |
| 7W141 | 10 | 46.47 | 236.44 | 62.2 | 47.9 | 56.5 | 313.8 | 5 | 82 | 8.5 | 9.0 | N | 205/14 | pb |
| 7W134 | 30 | 46.47 | 236.44 | 69.4 | 356.2 | 83.0 | 217.5 | 5 | 62 | 9.8 | 10.3 | N | 190/21 | cjf |
| 7W127 | 15 | 46.46 | 236.45 | 66.7 | 350.0 | 82.7 | 172.8 | 7 | 302 | 3.5 | 4.7 | N | 250/10 | cjf |
| 7W120 | 20 | 46.46 | 236.48 | 56.6 | 356.3 | 80.3 | 74.3 | 7 | 156 | 4.8 | 6.5 | N | 240/09 | cjf |
| 7W113 | 30 | 46.46 | 236.48 | 67.2 | 355.8 | 85.5 | 199.2 | 6 | 90 | 7.1 | 8.5 | N | 240/09 | cjf |
| 7W098 | 20 | 46.44 | 236.49 | −73.9 | 219.1 | −63.5 | 101.5 | 7 | 244 | 3.9 | 5.2 | R | 024/08 | sill |
| | | | | | *Domain 4* | | | | | | | | | |
| 8W094 | 20 | 46.57 | 236.62 | 56.2 | 330.6 | 66.1 | 133.0 | 7 | 94 | 6.3 | 8.4 | N | 230/25 | sill |
| 8W101 | 30 | 46.57 | 236.62 | 61.7 | 331.6 | 69.6 | 146.6 | 6 | 133 | 5.8 | 7.0 | N | 230/25 | pb |
| 8W108 | 30 | 46.57 | 236.62 | 66.4 | 346.1 | 80.4 | 165.4 | 5 | 81 | 8.5 | 9.0 | N | 230/29 | sill |
| 8W115 | 20 | 46.49 | 236.60 | 59.2 | 32.3 | 65.8 | 330.4 | 7 | 62 | 7.7 | 10.3 | N | 275/10 | sill |
| 8W122 | 30 | 46.49 | 236.60 | 69.4 | 3.6 | 83.0 | 254.7 | 6 | 32 | 12.0 | 14.4 | N | 275/10 | cjf |
| 9W108 | 20 | 46.48 | 236.53 | 71.7 | 3.0 | 79.8 | 245.9 | 7 | 260 | 3.8 | 5.0 | N | 279/10 | sill |
| 8W129 | 20 | 46.48 | 236.56 | 58.3 | 26.4 | 69.3 | 338.3 | 7 | 202 | 4.3 | 5.7 | N | 240/13 | cjf |
| 8W136 | 30 | 46.48 | 236.56 | 37.1 | 41.7 | 47.5 | 349.5 | 6 | 233 | 4.4 | 5.3 | N | 240/13 | sill |
| 9W101 | 20 | 46.47 | 236.53 | 58.3 | 357.3 | 32.3 | 72.3 | 6 | 100 | 6.7 | 8.1 | N | 260/10 | cjf |
| 9W094 | 20 | 46.47 | 236.53 | 44.0 | 39.9 | 52.3 | 345.8 | 6 | 49 | 9.7 | 11.6 | N | 299/13 | cjf |
| 8W143 | 30 | 46.47 | 236.53 | 54.4 | 11.2 | 75.7 | 16.4 | 7 | 176 | 4.6 | 6.1 | N | 270/19 | sill? |
| 9W043 | 20 | 46.47 | 236.58 | 56.0 | 9.4 | 77.9 | 18.0 | 7 | 187 | 4.4 | 5.9 | N | 200/12 | cjf |
| 9W029 | 20 | 46.47 | 236.55 | 63.7 | 19.1 | 76.7 | 324.5 | 7 | 152 | 4.9 | 6.6 | N | 225/21 | pb |
| 9W036 | 20 | 46.47 | 236.55 | 52.9 | 18.7 | 70.7 | 2.4 | 6 | 259 | 4.2 | 5.0 | N | 225/21 | pb? |
| 8W046 | 30 | 46.47 | 236.67 | 49.8 | 1.1 | 74.1 | 53.2 | 7 | 790 | 2.1 | 2.9 | N | 0/0 | pb |
| 8W053 | 30 | 46.47 | 236.67 | 46.3 | 6.7 | 70.4 | 38.7 | 6 | 83 | 7.4 | 8.9 | N | 0/0 | cjf |
| 8W039 | 20 | 46.47 | 236.67 | 60.5 | 2.3 | 84.7 | 37.6 | 7 | 30 | 11.2 | 14.8 | N | 120/17 | pb |
| | | | | | *Domain 5* | | | | | | | | | |
| 8W059 | 30 | 46.47 | 236.65 | 76.7 | 27.8 | 66.3 | 266.4 | 7 | 504 | 2.7 | 3.6 | N | 060/11 | sill |
| 8W066 | 20 | 46.47 | 236.65 | 71.9 | 60.7 | 52.3 | 288.0 | 5 | 61 | 9.9 | 10.4 | N | 060/11 | sill |
| 8W073 | 20 | 46.45 | 236.66 | 66.3 | 15.3 | 79.4 | 308.7 | 7 | 227 | 4.0 | 5.4 | N | 060/10 | sill |
| 8W150 | 30 | 46.44 | 236.66 | 82.6 | 35.6 | 57.4 | 252.4 | 5 | 168 | 5.9 | 6.3 | N | 135/26 | sill |
| 8W157 | 20 | 46.44 | 236.66 | 65.6 | 44.5 | 60.1 | 307.6 | 7 | 1351 | 1.6 | 2.2 | N | 128/27 | sill |
| 8W164 | 20 | 46.44 | 236.66 | 70.9 | 1.0 | 81.1 | 240.4 | 6 | 255 | 3.6 | 5.1 | N | 128/27 | sill |
| 8W171 | 30 | 46.44 | 236.65 | 52.9 | 26.4 | 66.1 | 350.2 | 6 | 53 | 9.3 | 11.2 | N | 145/19 | cjf |
| 8W080 | 20 | 46.44 | 236.69 | 68.5 | 21.0 | 75.3 | 297.8 | 7 | 99 | 6.1 | 8.2 | N | 135/20 | sill |
| 8W087 | 30 | 46.43 | 236.68 | −66.9 | 190.3 | −82.4 | 118.7 | 7 | 249 | 3.8 | 5.2 | R | 060/14 | sill |
| | | | | | *Domain 6* | | | | | | | | | |
| GL0B1 | 30 | 46.55 | 236.66 | −42.0 | 218.0 | −52.4 | 169.8 | 6 | 24 | 11.6 | 16.6 | R | 255/19 | cjf |
| GL0B2 | 30 | 46.55 | 236.69 | −76.0 | 248.0 | 49.6 | 99.2 | 6 | 128 | 5.1 | 7.2 | R | 245/16 | sill? |
| 9W022 | 20 | 46.51 | 236.77 | 62.5 | 43.0 | 59.7 | 317.1 | 7 | 212 | 4.2 | 5.6 | N | 310/30 | sill |
| 9W008 | 30 | 46.51 | 236.86 | 33.3 | 32.6 | 51.0 | 2.4 | 7 | 258 | 3.8 | 5.1 | N | 330/10 | pb |
| 8W004 | 30 | 46.47 | 236.73 | 70.2 | 20.7 | 74.8 | 288.5 | 7 | 84 | 6.6 | 8.9 | N | 295/14 | pb |
| 8W018 | 20 | 46.47 | 236.73 | 56.8 | 65.0 | 42.2 | 313.1 | 7 | 529 | 2.6 | 3.5 | N | 295/14 | cjf |
| 8W011 | 20 | 46.47 | 236.73 | 65.6 | 39.9 | 63.1 | 309.2 | 7 | 69 | 7.3 | 9.8 | N | 300/16 | cjf |
| 8W025 | 30 | 46.47 | 236.74 | 72.5 | 18.9 | 73.9 | 275.3 | 7 | 81 | 6.7 | 9.0 | N | 0/0 | pb |
| 8W032 | 30 | 46.46 | 236.73 | 49.2 | 37.1 | 57.0 | 343.3 | 7 | 21 | 13.3 | 17.6 | N | 292/05 | pb |
| 8W178 | 20 | 46.40 | 236.75 | 60.7 | 51.7 | 53.2 | 314.8 | 7 | 61 | 7.8 | 10.5 | N | 020/05 | cjf |

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TABLE 1. (continued)

| Site | Demagnetization Field, mT | Site Location | | Remanence Direction | | VGP Position | | N | Statistics | | | Polarity | Strike and Dip | Rock Type |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Latitude | Longitude | Inclination | Declination | Latitude | Longitude | | $k$ | $\alpha_{95}$ | ASD | | | |
| | | | | | *Domain 6 (continued)* | | | | | | | | | |
| 8W185 | 20 | 46.39 | 236.74 | 60.1 | 42.9 | 58.9 | 320.9 | 6 | 192 | 4.8 | 5.9 | N | 070/03 | sill |
| 9W015 | 30 | 46.37 | 236.83 | 60.4 | 51.5 | 53.2 | 315.5 | 7 | 603 | 2.5 | 3.3 | N | 340/13 | sill |
| | | | | | *Domain 7* | | | | | | | | | |
| 7W023 | 30 | 46.30 | 236.11 | −34.5 | 246.3 | −29.8 | 149.4 | 4 | 35 | 15.7 | 13.8 | R | 010/35 | pb |
| 7W027 | 30 | 46.30 | 236.11 | −49.3 | 226.7 | −50.6 | 153.5 | 5 | 322 | 4.3 | 5.6 | R | 000/30 | pb |
| 0W008 | 30 | 46.27 | 236.14 | 43.7 | 36.0 | 54.7 | 349.5 | 7 | 54 | 8.3 | 11.1 | N | 340/25 | pb |
| 7W001 | 30 | 46.28 | 235.96 | −56.9 | 250.2 | −38.7 | 129.1 | 7 | 59 | 7.9 | 10.6 | R | 232/28 | pb |
| 0W001* | 30 | 46.27 | 236.14 | 19.1 | 64.1 | 24.9 | 338.4 | 7 | 40 | 9.6 | 12.6 | N | 307/27 | pb |
| 7W009* | 50 | 46.27 | 236.08 | −2.5 | 253.6 | −12.2 | 157.2 | 7 | 54 | 8.3 | 11.1 | R | 325/30 | pb |
| 7W017* | 50 | 46.27 | 236.08 | −2.5 | 240.7 | −20.7 | 167.3 | 6 | 19 | 15.7 | 17.8 | R | 310/35 | pb |

$N$ is the number of cores per site; $k$ is the precision parameter of *Fisher* [1953]; $\alpha_{95}$ is the radius of the cone of 95% confidence about the mean, in degrees; ASD is angular standard deviation, in degrees; strike azimuth is left-handed direction when facing downdip; cjf is columnar jointed interior of submarine flow; pb is pillow basalt; sill is columnar jointed to massive basalt or gabbro sill.

*Site not used in calculation of formation mean; site descriptions in the work of *Wells* [1982].

the same cores (Figure 5). Temperatures between 300° and 450°C produced good grouping of magnetization directions within a site, and maximum blocking temperatures ranged between 450° and 550°C. A few sites in areas of hydrothermal alteration were characterized by weak NRM's ($10^{-3}$ A/m) and magnetization directions stable above 600°C, indicating hematite as the carrier of remanence. Two of these sites were rejected because the very stable NRM directions failed to converge toward a common direction with thermal demagnetization.

Most of the thermally demagnetized samples displayed moderately high blocking temperatures which, assuming that they initially possessed the 180° to 200°C Curie temperatures typical of fresh seafloor basalts [see *Grommé et al.*, 1979], indicates that oxidation of the primary titanomagnetites has taken place. The ubiquitous occurrence of celadonite or saponite clays as vesicle fillings or as alteration products of interstitial basaltic glass suggests that low-temperature interaction with seawater has affected the Crescent submarine basalts. A rather thorough low-temperature oxidation or maghemitization of the titanomagnetite could explain in part the higher blocking temperatures and low intensities of magnetization in the pillow basalts [*Grommé et al.*, 1979; *Prévot et al.*, 1981].

### *Reliability of Paleomagnetic Results*

Although the Crescent Formation undoubtedly has undergone low-temperature oxidation and local hydrothermal alteration, there are several lines of evidence that suggest that the magnetization was acquired in the direction of the geomagnetic field at essentially the time the rocks formed and that the mean magnetization direction approximates the direction of a geocentric axial dipole field.

*Fold test.* Site locations in the Crescent Formation are widely distributed throughout several complexly folded uplifts. If we rotate the magnetization directions about the local strike of observed bedding to the assumed paleohorizontal, we produce a decrease in scatter of magnetization directions significant at the 99% confidence interval (Figure 7 and Table 2). This positive fold test shows that the age of magnetization predates the folding, which began during the middle Eocene.

*Polarity test.* Both normal and reversed polarities are observed in the Crescent Formation, and their respective mean directions are essentially antipolar (Table 2). This shows that we have averaged out short-term secular variation and are dealing with an average direction which provides an estimate of the axial dipole field during the time of extrusion and that systematic secondary components are not a problem. Although scatter in the reversed directions is greater than that of the normal directions and suggests the possibility of retention of small components of secondary magnetization, the antipolarity of the mean normal and reversed directions indicates that any such secondary components are random.

*Consistency between rock types.* *Johnson and Merrill* [1973], *Johnson and Hall* [1978], and *Prévot et al.* [1981] have suggested that submarine flows acquire a chemical remanent magnetization (CRM) during low-temperature oxidation. Coarse-grained basalts in flow interiors and sills, if they escaped high-temperature oxidation, would be subject to slow oxidation by "seafloor weathering." The multidomain titanomagnetite grains in these basalts might be susceptible to changes in magnetization direction by CRM, either by reorganization of domain walls or by trapping of easily acquired viscous components as coercivity increases with increasing oxidation [*Marshall*, 1978]. In contrast, the pillow basalts would be less susceptible to CRM-related direction changes because of their fine grain size and relatively rapid oxidation relative to long-term processes affecting the direction of the ambient field (e.g., tectonic transport or rotation).

A comparison of cleaned magnetization directions for different basalt types from similar structural blocks, however, shows that the mean direction for pillow basalts and coarse-grained flow interiors are essentially the same within statistical limits (Table 3). The magnetization direction from the sills in any one block also is consistent with the direction of pillow basalt and flow interior means from that block. Although the preci-

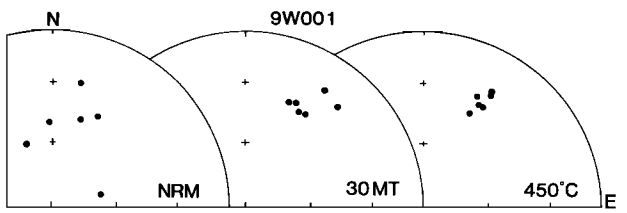



Fig. 5. Lower hemisphere equal area plot of sample magnetization directions for site 9W001 in pillow basalt flow of the Crescent Formation. Compare scatter of NRM directions with directions after alternating field demagnetization at 30 mT or thermal demagnetization at 450°C.

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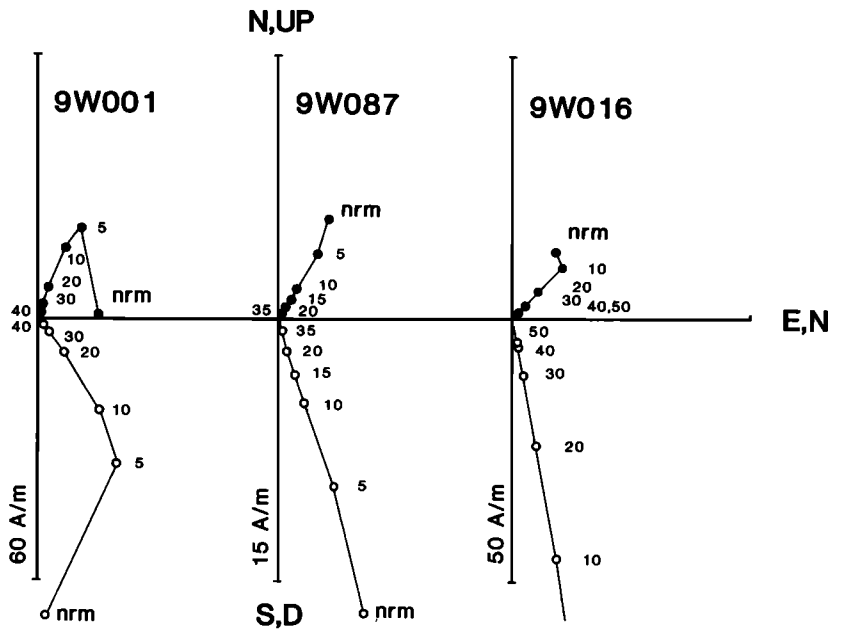



Fig. 6. Vector demagnetization diagrams for pilot AF demagnetization of samples from sites 9W001, 9W015, and 9W087 in pillow basalt flows of the Crescent Formation, before tilt correction. Solid symbols represent projection of magnetization vector into the horizontal plane; open symbols represent projection into the N-S vertical plane.

sion parameter ($k$) for the sills decreases slightly upon unfolding of the tilted beds, it is statistically insignificant given the small number of sites in the sills. The results from our samples in the Willapa Hills are also generally consistent with the results of *Globerman et al.* [1982] from subaerial basalts of the Crescent Formation in the Black Hills, 60 km to the north. Thus it seems very unlikely that the characteristic directions of our samples have been significantly affected by CRM acquired during low-temperature oxidation. Overall, the Crescent Formation magnetization directions from the Willapa Hills appear to be reliable and useful for tectonic interpretations.

*Discussion*

The mean Crescent Formation pole is rotated 34° ± 7° east of the Eocene reference pole for stable North America [*Diehl et al.*, 1983] and indicates a barely significant northward latitude shift (5° ± 4.8°) (Table 2). Although no other uplifts of Coast Range pillow basalt basement show a significant latitude shift [*Simpson and Cox*, 1977; *Globerman et al.*, 1982; *Beck and Engebretson*, 1982], slight flattening of inclinations is ubiquitous and may indicate small (<300 km?) northward translation [*Bates et al.*, 1981]. Alternatively, the slight inclination flattening in the Willapa Hills may result from tilt correcting predominantly north dipping asymmetric folds containing minor postfolding magnetization or small initial dips (<5°) to the north. The declinations are similar to those from other studies in Eocene and Oligocene rocks of the Washington Coast Range and western Cascade Range [*Beck and Burr*, 1979; *Bates et al.*, 1981; *Globerman et al.*, 1982] and might suggest coherent clockwise rotation of most of Washington west of the Cascade Range (Figure 8). However, when the declinations are plotted on a geologic map of southwest Washington, a pattern emerges that suggests subdivision of the Crescent uplift into separate blocks (Figure 9).

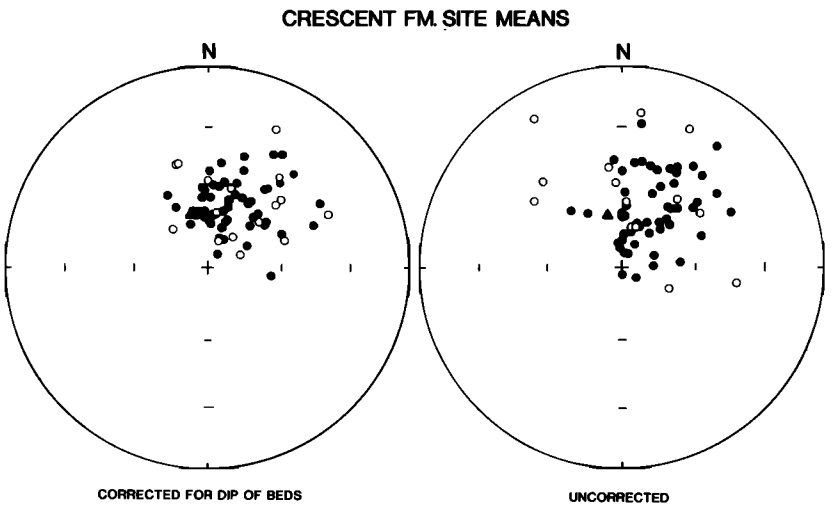



Fig. 7. Equal area plot of site mean magnetization directions before and after tilt correction for the middle Eocene Crescent Formation, Willapa Hills, southwest Washington. Closed symbols are normal polarity lower hemisphere; open symbols are reversed polarity directions inverted through origin for comparison. Triangle is expected Eocene magnetization direction for a stable southwest Washington.

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TABLE 2. Paleomagnetic Data From the Crescent Formation, Southwest Washington, Formation Mean and Domain Means

| | Pole Position | | Pole Statistics | | | Direction Statistics | | | | Remanence Direction | | Latitude Shift | Rotation |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Latitude | Longitude | $A_{95}$ | $K$ | $N$ | $\alpha_{95}$ | $k$ | ASD | $\overline{\text{ASD}}$ | Inclination | Declination | $L \pm \Delta L$ | $R \pm \Delta R$ |
| Formation mean tc | 73.3 | 326.4 | 5.3 | 11 | 70 | 4.1 | 19 | 19.0 | 8.9 | 61.0 | 024.6 | $5.0 \pm 4.8$ | $33.6 \pm 6.7$ |
| Formation mean unc | 73.0 | 315.9 | 6.5 | 8 | | 5.0 | 13 | 23.2 | | 62.3 | 024.5 | | |
| Normal | 73.4 | 325.5 | 5.6 | 13 | 56 | 4.2 | 21 | 17.7 | 8.4 | 61.4 | 024.9 | | |
| Reversed* | 73.0 | 330.3 | 15.8 | 8 | 14 | 12.3 | 11 | 24.3 | 10.8 | 59.1 | 023.8 | | |
| Domain 1 tc | 62.8 | 329.6 | 29.6 | 8 | 5 | 23.1 | 12 | 23.6 | 12.2 | 56.1 | 038.6 | $10.1 \pm 23.2$ | $46.0 \pm 30.5$ |
| Domain 1 unc | 78.7 | 057.2 | 40.3 | 5 | | 34.5 | 6 | 34.0 | | 51.5 | 356.6 | | |
| Domain 2 tc | 80.2 | 358.9 | 18.1 | 7 | 11 | 13.5 | 12 | 23.2 | 10.6 | 58.0 | 011.3 | $8.3 \pm 14.3$ | $21.0 \pm 19.1$ |
| Domain 2 unc | 67.3 | 021.3 | 18.0 | 7 | | 15.8 | 9 | 27.0 | | 42.7 | 015.3 | | |
| Domain 3 tc | 82.2 | 303.1 | 10.6 | 19 | 11 | 7.6 | 37 | 13.4 | 8.8 | 65.4 | 011.4 | $-0.1 \pm 8.6$ | $21.0 \pm 13.2$ |
| Domain 3 unc | 74.0 | 268.0 | 16.0 | 9 | | 10.2 | 21 | 17.7 | | 72.5 | 017.3 | | |
| Domain 4 tc | 80.7 | 006.0 | 8.4 | 19 | 17 | 6.5 | 32 | 14.5 | 7.9 | 58.5 | 010.5 | $8.8 \pm 7.0$ | $19.5 \pm 9.3$ |
| Domain 4 unc | 70.9 | 296.2 | 10.7 | 12 | | 7.7 | 23 | 17.1 | | 67.3 | 027.6 | | |
| Domain 5 tc | 71.7 | 289.7 | 10.1 | 27 | 9 | 6.6 | 63 | 10.3 | 6.4 | 70.0 | 025.8 | $-5.9 \pm 8.2$ | $36.0 \pm 14.2$ |
| Domain 5 unc | 74.2 | 344.6 | 11.7 | 20 | | 9.4 | 31 | 14.6 | | 57.8 | 020.6 | | |
| Domain 6 tc | 59.8 | 319.0 | 9.0 | 24 | 12 | 7.6 | 33 | 14.3 | 8.6 | 59.8 | 042.0 | $6.8 \pm 7.4$ | $52.4 \pm 10.2$ |
| Domain 6 unc | 59.4 | 299.1 | 14.7 | 10 | | 9.8 | 21 | 17.9 | | 66.4 | 046.0 | | |
| Domain 7 tc | 44.3 | 328.8 | 18.7 | 25 | 4 | 16.7 | 31 | 18.1 | 10.8 | 47.0 | 054.5 | $19.9 \pm 14.8$ | $65.0 \pm 17.1$ |
| Domain 7 unc | 54.8 | 316.7 | 32.5 | 7 | | 23.3 | 12 | 23.8 | | 59.4 | 052.7 | | |
| Domain 234 tc | 81.9 | 349.2 | 6.5 | 14 | 39 | 4.9 | 23 | 17.0 | 8.9 | 60.4 | 011.0 | $6.1 \pm 5.6$ | $20.3 \pm 10.8$ |
| Domain 234 unc | 76.1 | 313.4 | 8.7 | 8 | | 6.8 | 12 | 23.4 | | 62.4 | 020.6 | | |
| Eocene pole North America† | 82.8 | 170.4 | 3.0 | 16.1 | | | | | | 66.5 | 349.9‡ | | |

Abbreviations and variables: tc, corrected for dip of beds; unc, uncorrected for dip of beds; $A_{95}$ and $\alpha_{95}$, radius of cone of 95% confidence about mean in degrees; $K$ and $k$, precision parameter; ASD, angular standard deviation in degrees; $\overline{\text{ASD}}$, average flow ASD; $L$, latitude shift in degrees; positive $L$ value, northward shift; negative $L$ value, southward shift; $R$, rotation in degrees calculated from poles; positive $R$ value, clockwise rotation; $\Delta L$, error limits on latitude shift; $\Delta R$, error limits on rotation; uncertainties calculated by method of *Demarest* [1983].

*Reversed direction inverted through origin for comparison.

†Eocene pole calculated from *Diehl et al.* [1983].

‡Expected direction for Willapa Hills, southwest Washington.

As a result of our geologic and paleomagnetic studies, we have subdivided the Crescent Formation basement into seven major structural domains (compare Figures 3 and 9). Those areas where easterly declinations are more pronounced (as in the eastern Willapa Hills uplift or near the mouth of the Columbia River) are geographically distinct, structurally separated from the rest of the Willapa Hills uplift, and form locally significant subgroups within the Crescent paleomagnetic data set. The boundaries between most domains are major structural breaks such as the Naselle River fault zone between domain 7 and the other domains, or the Willapa River fault zone between domains 2 and 3. In other places, the boundary marks an abrupt change in structural trend or style not known to be associated with a throughgoing major fault.

The discordance of paleomagnetic directions between most of the domains is probably the result of tectonic rotation, because the paleomagnetic domains are concident with structural domains determined by geologic mapping and are separated from one another by major fault zones. Secular variation is an unlikely cause for the dispersion in declination for two reasons: (1) most domains contain both normally and reversely polarized rocks whose directions are nearly antiparallel (with the exception of the northernmost domain, discussed later) and (2) the overall scatter in the entire Crescent data set is much larger than one would expect from secular variation alone. When the Crescent is subdivided into domains, the scatter within most domains (Table 2) decreases and approaches values one might expect from secular variation at this latitude [*Cox*, 1970; *McElhinny and Merrill*, 1975].

Using the structural boundaries illustrated in Figure 9, we have subdivided the Crescent results (Figure 10) and calculated average poles and magnetization directions for each domain. Rotations and apparent latitude shift were determined by comparing the observed pole to the expected Eocene pole from *Diehl et al.* [1983] (Table 2). The mean clockwise rotation for each domain is shown in Figure 11. A consistent rotation of about 21° is characteristic of a "core area" composed of domains 2, 3, and 4 while the surrounding domains 1, 5, 6, and 7 show greater rotations of about 45° to 65° clock-

TABLE 3. Paleomagnetic Data From the Crescent Formation, Southwest Washington, Rock Type Means

| | Pole Position | | Pole Statistics | | | Direction Statistics | | | | Remanence Direction | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Domain 234 Rock Type | Latitude | Longitude | $A_{95}$ | $K$ | $N$ | $\alpha_{95}$ | $k$ | ASD | $\overline{\text{ASD}}$ | Inclination | Declination |
| Flow interiors, tc | 83.8 | 008.1 | 8.4 | 24 | 14 | 6.2 | 42 | 12.5 | 8.6 | 60.4 | 007.4 |
| Flow interiors, unc | 83.4 | 305.1 | 13.2 | 10 | | 10.1 | 16 | 20.2 | | 64.5 | 007.9 |
| Pillow basalt, tc | 79.4 | 355.1 | 11.8 | 10 | 17 | 8.9 | 17 | 19.9 | 10.3 | 58.3 | 012.5 |
| Pillow basalt, unc | 71.0 | 339.5 | 15.0 | 7 | | 12.0 | 10 | 26.2 | | 54.2 | 024.0 |
| Sills, tc | 80.8 | 309.3 | 16.8 | 12 | 8 | 12.6 | 20 | 18.1 | 7.2 | 64.5 | 014.4 |
| Sills, unc | 65.4 | 276.7 | 14.8 | 15 | | 10.8 | 27 | 15.6 | | 73.2 | 036.4 |

See Table 2 for explanation of symbols.

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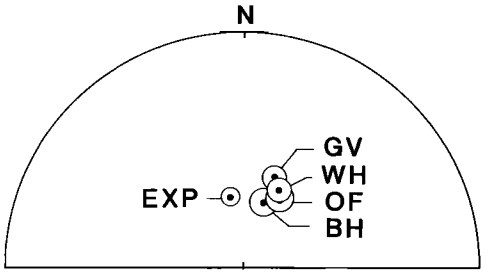



Fig. 8. Equal area plot of mean magnetization direction for the Crescent Formation in the Willapa Hills (WH) and $\alpha_{95}$ compared to results of other paleomagnetic studies in southwest Washington; BH, Crescent Formation in the Black Hills [*Globerman et al.*, 1982]; GV, Goble Volcanics [*Beck and Burr*, 1979]; OF, Ohanapecosh Formation [*Bates et al.*, 1980]; EXP, expected direction from Eocene North American reference pole of *Diehl et al.* [1983].

wise. Within each of the domains 3, 4, 5, and 6, the paleomagnetic directions are internally consistent, with $k$ ranging from 32 to 63. Scatter in the other domains is considerably greater and probably has several different causes.

In domain 2 the directional scatter ($k = 11$) is much greater than the scatter in nearby domains 3 and 4 ($k = 37$ and 32). The average rotation of all three domains is essentially identical, but domain 2 is structurally much more complicated than domains 3 and 4 (Figure 3). This suggests that coherent rotation of all three areas occurred, with domain 2 suffering more internal disruption because of its outboard position. In domain 1, however, the strongly discordant directions can be related to separate fault-bounded structural subdomains which are characterized by diverging structural geometries (Figures 3 and 9). Thus the average rotation of domain 1 is probably not very useful for further tectonic analysis.

In contrast, magnetization directions from domain 7 are scattered along inclinations rather than declinations (Figure 10). Both reversed and normal polarities are represented, and they are roughly antiparallel, although some are quite shallow in inclination (average 42°, see Table 1). Considering that the stratigraphy of domain 7 is essentially identical to that of the other domains, it is unrealistic to suggest that domain 7 has undergone large northward transport with respect to the rest of the Coast Range basement. More likely possibilities are that some domain 7 flow magnetizations have recorded either a polarity transition or a partial remagnetization that occurred prior to folding. If we discard the three shallow directions ($I < 30°$), the domain mean direction (Table 2) suggests about 65° of clockwise rotation, but $N$ is small and the uncertainties are large. Thus the results from domain 7 are not sufficient to form the basis for major tectonic arguments. However, tectonic rotation is the most likely explanation for the eastward deflection of directions in domain 7, as it is for all of the other domains.

To summarize, the domains with useful mean directions are domain 2 with 21° ± 19° clockwise rotation, domain 3 with 21° ± 13°, domain 4 with 20° ± 9°, domain 5 with 36° ± 14°, domain 6 with 52° ± 10°, and perhaps domain 7 with 65° ± 17° clockwise rotation. We can combine the similar domains 2, 3, and 4 into one domain (234) and then compare the rotation of this large central domain (20° ± 7°) with the rotation of other domains. When compared in this way, domains 5, 6, and 7 are significantly more clockwise rotated than 234, the difference being 16° ± 15°, 32° ± 12°, and 44° ± 18°, respectively.

The data indicate that differential rotations in the Crescent Formation basement have occurred on several scales. Widely scattered directions from domains 1 and 2 suggest differential rotation of small fault-bounded blocks a few kilometers on a side while the internally coherent rotation of domains 3 and 4, when compared to domains 6 and 7, suggest differential rotations of larger blocks up to 30 km on a side bounded by major zones of deformation.

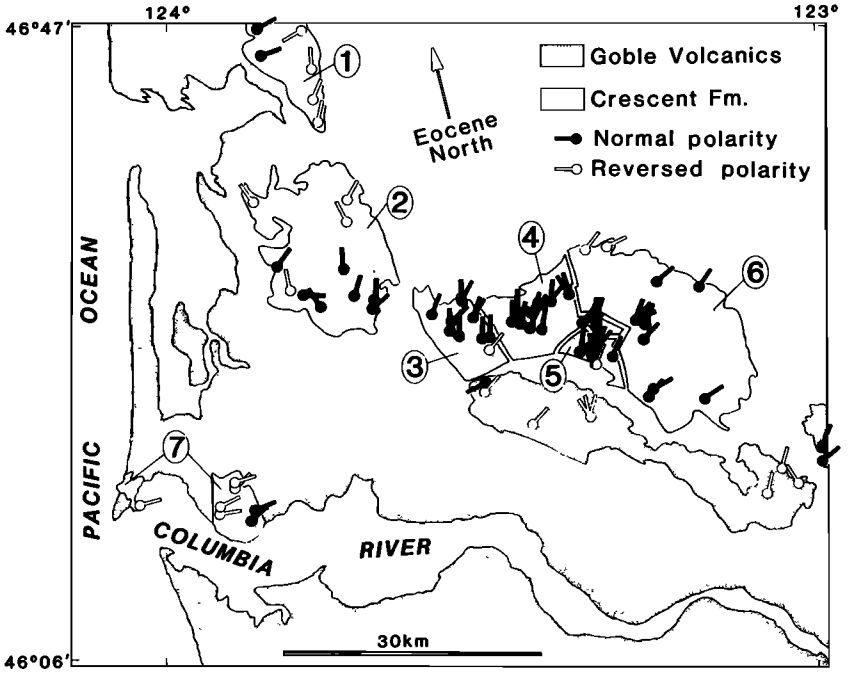



Fig. 9. Paleomagnetic site localities in the Crescent Formation and Goble Volcanics, southwest Washington. Azimuth of magnetization direction shown by arrows; solid circle, normal polarity; open circle, reversed polarity; azimuth inverted through origin.

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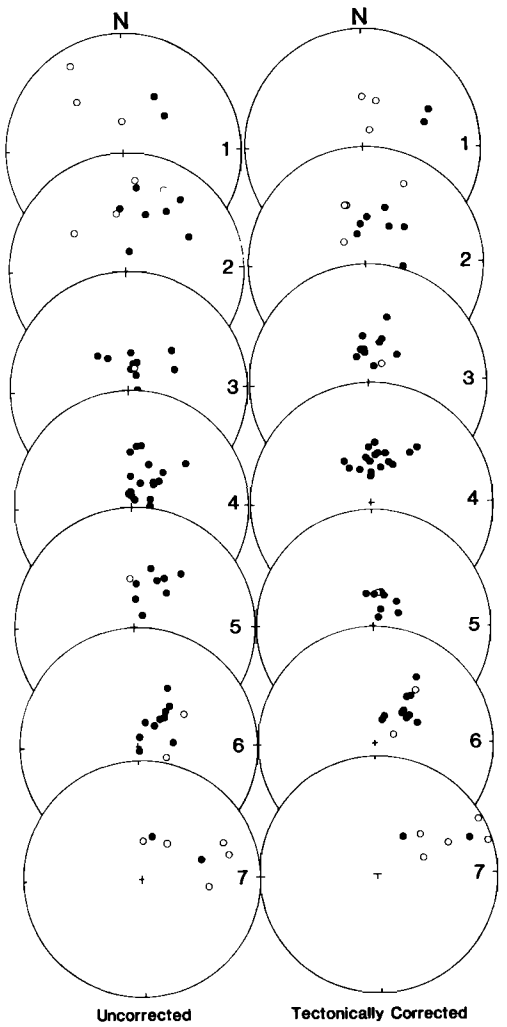



Fig. 10. Equal area plots of site mean magnetization directions before and after tectonic correction for tectonic domains in the Crescent Formation. Solid symbols, normal polarity lower hemisphere; open symbols, reversed polarity inverted through origin for comparison.

## 3. STRUCTURAL ANALYSIS OF THE CRESCENT FORMATION

In this section we consider two lines of evidence that demonstrate a relationship between geologic structure and tectonic rotations in the Crescent Formation: (1) zones of more intense deformation bounding differentially rotated domains and (2) fold axis orientations within domains that are correlated with the amount of rotation.

### *Domain Boundary Structures*

The margins of our chosen structural and paleomagnetic domains are more intensely deformed and commonly marked by vertical to overturned isoclinal folds and accompanying major shear zones. These zones have apparently taken up the motion between differentially rotating blocks through a combination of folding, thrusting, and strike-slip motion. The best exposed tectonic boundary is in the eastern Willapa Hills, where it was predicted by a marked change in declination in our early paleomagnetic sampling. Subsequent field mapping in the area confirmed the existence of the eastern Willapa Hills (EWH) fault zone which separates domains 4 and 5 on the west from domain 6 on the east [*Wells*, 1981]. It consists of a major compressional and dextral shear zone marked by vertical to slightly overturned, north-northwest trending isoclinal folds and throughgoing north-northwest dextral faults (Figure 12). Although sinistral conjugate faults also are present, the dextral faults predominate, and everywhere that fault planes are exposed the grooves and slickensides are nearly horizontal. The EWH fault zone represents a zone of convergence and dextral displacement between the EWH block (domain 6) and less rotated blocks to the west, with the folding taking up a minimum of 4 km of compression between the two blocks. The most intense folding and shearing occurs south of the Thrash Creek fault system (Figure 12). North of the Thrash Creek faults, the compressive zone is replaced by an ill-defined broad zone of widely spaced faults of unknown displacement. Deformation in the EWH suggests that a pivot point for the relative rotation lies somewhere near the north end of the zone of major compression where the tectonic style changes abruptly ("P" in Figure 12). Using "P" as a pivot point requires a maximum of about 8 km of convergence along the EWH fault zone to account for the 32° differential rotation between the two blocks. This amount is in reasonable accord with the minimum 4-km convergence determined from unfolding the observed structures and would be easily attainable by minor thrusting at depth along the boundary faults.

### *Fold Axes and Rotations*

In this analysis, we divided the large paleomagnetic domains of the Crescent uplift into 13 structural subdomains to more accurately define the fold axes. This effectively segregates the paleomagnetic data into the large "A" subdomains (compare Figures 3, 11, and 13). To look for a quantitative relationship between folding and rotation, we plotted 783 poles to bedding for the 13 domains and calculated best fit great circles to the data to define a mean fold axis for each domain (Figure 13 and Table 4).

The domain fold axes are clustered roughly into 3 groups, a northwest, northeast, and an east-west group. In general, the plunge of the fold axes is small (0 to 15°), and the spread of fold axes around a vertical axis of rotation is much greater than the variation in plunge. However, several factors cause moderate to large uncertainties in the data: (1) incomplete sampling of domains with poor exposure (e.g., domain 1); (2) the existence of small subdomains with differing structural trends within some large domains (e.g., domains 2 and 7); and (3) gentle dips and broad, poorly defined fold culminations in some domains (e.g., domain 2). Although uncertainties are large, the mean fold axis, in most cases, is a good approximation to the average trend of fold axes in each domain when the mean is compared to local trends determined from geologic mapping (Figure 3).

The A domains (Figure 13) with their northwest trending folds are the most significant in terms of total area affected by deformation. They record a major folding event of late middle Eocene age which probably affected most of southwest Washington. Other domains which contain northeast and east-west folds (generally the "B" groups in Figure 13) cover much smaller areas and are generally restricted to the margins of the basement uplifts. Some of these domains may represent local areas of greater tectonic rotation (as in 6B?), but we do not have the paleomagnetic data to test this idea. Marginal deformation of block uplifts may in part be of Miocene age and may contribute some "noise" to the overall domain picture, but the general discordance between Crescent Formation structures and those in younger sedimentary rocks makes it easy to define early formed structures in the Crescent Formation.

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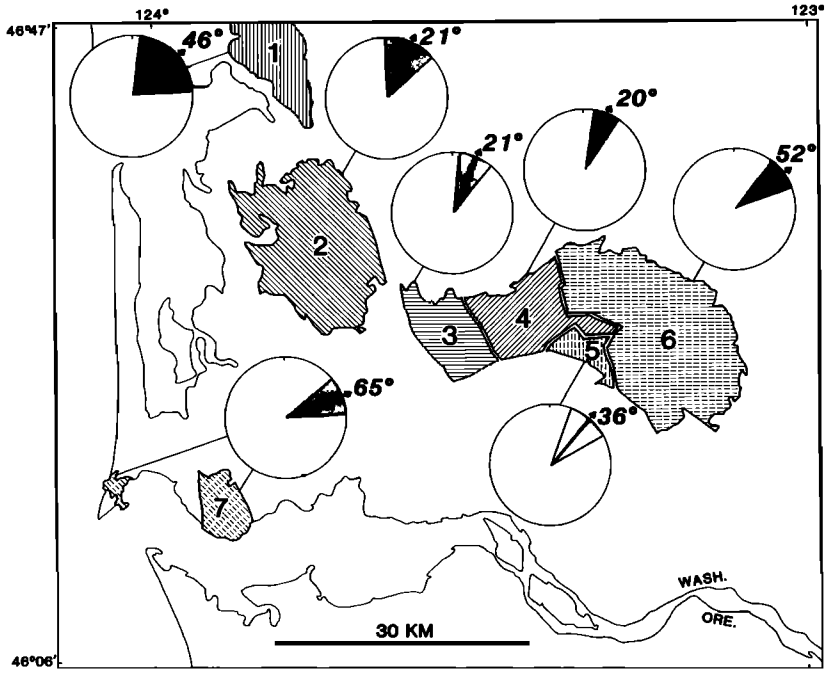



Fig. 11. Rotations of paleomagnetic domains (Table 1, Figure 10) in the Crescent Formation. Arrow shows rotation calculated relative to Eocene reference pole and plotted with respect to geographic north; 95% confidence limits on rotation shown by shaded area.

The horizontal dispersion of fold axes among the A domains may be related to tectonic rotation of the domains. Ideally, if the fold axes were formed prior to rotation, the angular separation of domain fold axes should correspond directly to the differential rotations between those domains. A plot of rotation versus azimuth of fold axes for domains with statistically significant paleomagnetic data suggests a positive correlation between rotation and fold azimuth for domains 3A, 4A, 6A, and 7A (Figure 14). Domain 2 does not fall on the trend established by other domains, probably because of its structural complexity. The large scatter in both paleomagnetic directions and poles to bedding indicates small block rotations have occurred in domain 2, as discussed earlier.

Although the large uncertainties do not permit rigorous analysis, the correlation between folding and rotation suggests that folding in the Crescent basement began prior to differential tectonic rotation. This folding is not likely a product of oceanic tectonics prior to accretion; more likely it is the result

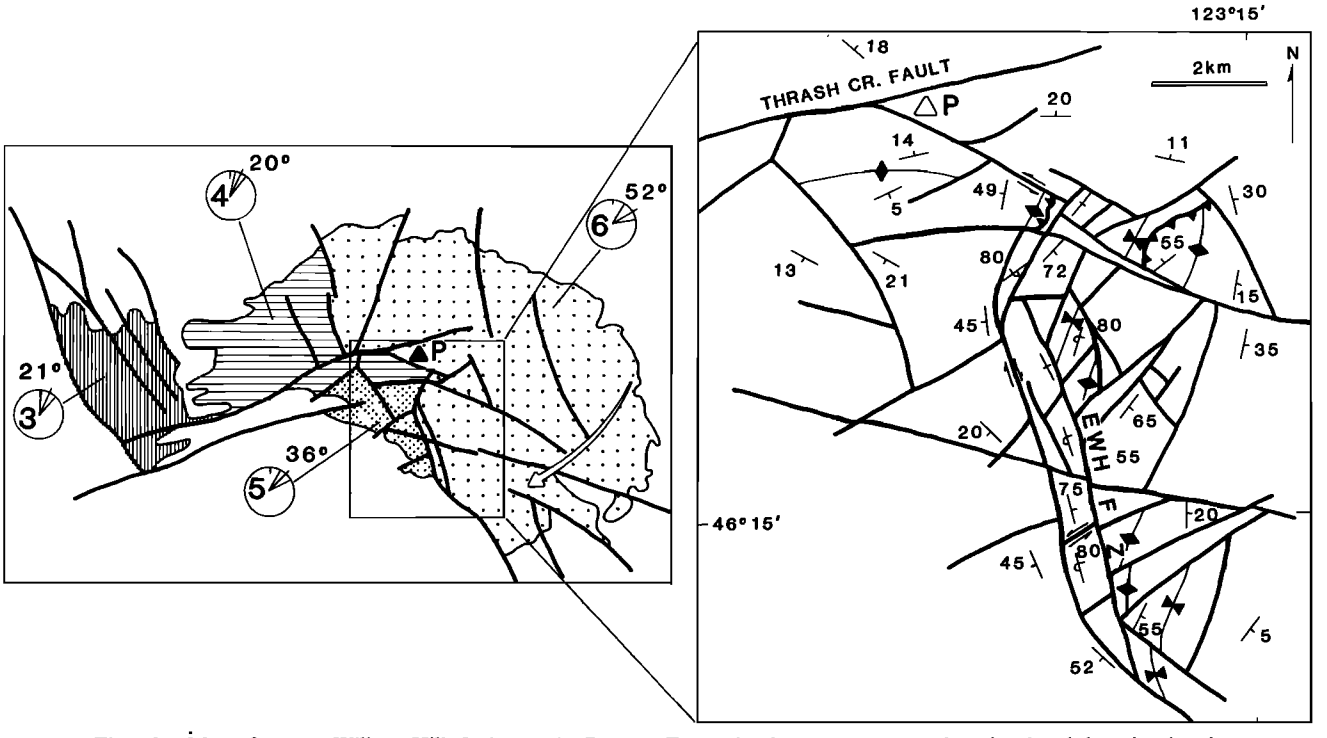



Fig. 12. Map of eastern Willapa Hills fault zone in Crescent Formation between structure domains 6 and domains 4 and 5; rotations as in Figure 11. P is possible pivot point for domain 6 rotation; see text for discussion.

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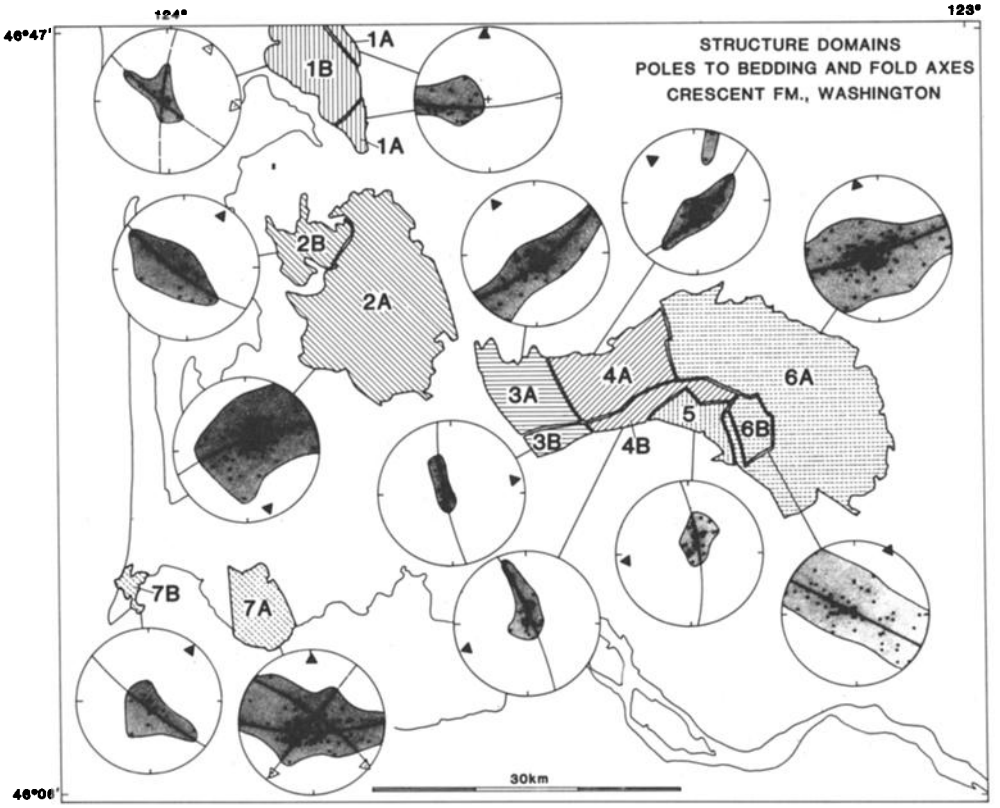



Fig. 13. Structural domains in the Crescent Formation. Equal area plots show poles to bedding (dots) and best fit fold axis (solid triangle) for each domain. Open triangles are estimated fold axes for apparent subdomains within some domains.

of deformation against the buttress of the North American continent. Thus it would appear that most of the differential rotation of the domains occurred after emplacement along the continental margin rather than during fragmentation of an oceanic plate prior to accretion.

### *Timing of Differential Rotations*

The timing of the differential rotations in the Cresent Formation is constrained by the age of the oldest rocks unaffected by the differential rotations. The EWH fault zone cuts the early and middle Eocene Crescent Formation as well as the Eocene gabbros that intrude the Crescent basalts and the overlying middle Eocene siltstones. Radiometric ages from the gabbros give early and middle Eocene ages (R. Wells and M. J. Heatherington, unpublished data, 1980), but they are not yet definitive. The lack of similar gabbros in strata younger than middle Eocene is consistent with a middle Eocene age for the gabbros (about 44 to 48 m.y.). Because the gabbros are differentially rotated, we can establish a maximum age of early middle Eocene for the differential rotation. A minimum age limit on the differential rotations is afforded by the upper middle and upper Eocene strata of the Cowlitz Formation and the Goble Volcanics which overlap the boundary structures in the Crescent Formation with an angular unconformity [*Wells*, 1981]. The prominent north-northwest isoclinal folding and extensive shearing typical of the EWH fault zone are absent in these younger rocks, although relatively mild en echelon faulting does occur on trend (Figure 2), probably as a result of continued activation of the older buried structures. For additional information on the timing of differential rotations, we collected paleomagnetic samples across the probable subsurface extension of the EWH fault zone in the unconformably overlying Eocene Goble Volcanics.

TABLE 4. Best Fit Fold Axes for Structure Domains in the Crescent Formation

| Domain | Fold Axis | $A_{95}$ | $B_{95}$ | $\lambda_{max}$ | $\lambda_{min}$ | $\lambda_{int}$ | $N$ |
|---|---|---|---|---|---|---|---|
| 1A | −9.9/177 | 56.0 | 13.1 | 17.1 | 0.839 | 2.06 | 20 |
| 1B | −6.6/236.8 | 50.6 | 14.2 | 9.94 | 0.561 | 1.50 | 14 |
| 2A | 9.1/160 | 57.5 | 19.2 | 87.4 | 7.52 | 18.1 | 165 |
| 2B | −10.9/217.5 | 46.7 | 23.4 | 8.44 | 0.914 | 2.64 | 12 |
| 3A | 13.1/141.6 | 36.5 | 14.5 | 33.7 | 1.71 | 6.55 | 67 |
| 3B | −8.0/258.5 | 15.0 | 5.5 | 13.2 | 0.105 | 1.67 | 16 |
| 4A | 13.4/315.2 | 36.3 | 7.4 | 55.6 | 0.912 | 3.52 | 58 |
| 4B | −5.9/68.3 | 38.2 | 13.2 | 22.5 | 0.983 | 3.55 | 26 |
| 5 | −12.6/83.7 | 90 | 11.9 | 22.6 | 0.925 | 1.52 | 36 |
| 6A | −7.2/162.3 | 90 | 14.3 | 146 | 8.07 | 14.1 | 167 |
| 6B | 0.1/207.9 | 31.6 | 18.7 | 43.7 | 4.09 | 19.2 | 58 |
| 6C | 0.0/162.9 | 12.1 | 6.2 | 14.8 | 0.169 | 4.03 | 28 |
| 7A | −11.3/181 | 50.1 | 19.5 | 78.2 | 6.7 | 18.1 | 102 |
| 7B | −7.3/220.4 | 58.3 | 13.6 | 12.7 | 0.668 | 1.59 | 15 |

Plunge and azimuth of fold axis are given in degrees; negative values are upper hemisphere; $A_{95}$ and $B_{95}$ are major and minor axes of error ellipse, respectively, in degrees; $\lambda_{max}$, $\lambda_{min}$, and $\lambda_{int}$ are eigenvalues for best fit fold axes and great circles; $N$ is the number of poles to bedding; 6A and 6C combined in Figure 13.

## 4. PALEOMAGNETISM OF THE GOBLE VOLCANICS

The Goble Volcanics (Figure 2) consists of a gently south dipping sequence of thin subaerial basalt flows, flow breccias, and thin interbeds of sandstone, siltstone, and weathered basaltic debris [*Livingston*, 1966; *Wells*, 1981]. In the Washington Coast Range north of Cathlamet, these volcanic rocks overlie upper Eocene sandstones of the Cowlitz Formation and are in turn overlain by upper Eocene deep-water silt-

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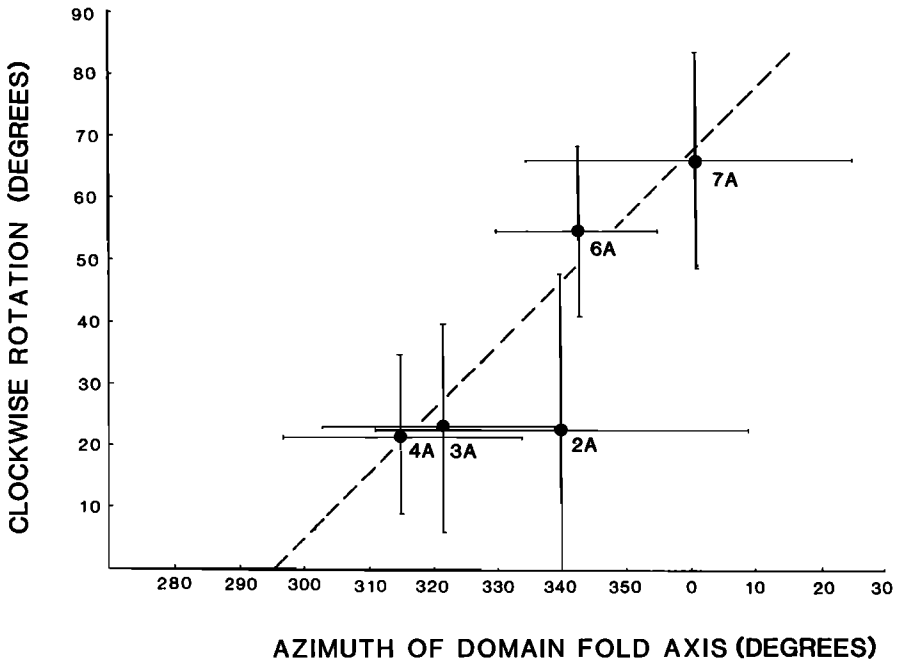



Fig. 14. Rotation versus fold axis azimuth for the major structural domains in the Crescent Formation, as labeled in Figure 13. Error limits on fold axis azimuth determined from major axis of error ellipse around mean. Positive correlation between rotation and fold axis azimuth indicates rotation occurred after early folding possibly related to accretion of Coast Range. Dashed line shows ideal 1:1 correspondence.

stones [*Wells*, 1981]. Potassium-argon ages of 39 and 32 m.y. have been determined for two flows in the Goble Volcanics above the Cowlitz Formation (R. E. Wells, M. J. Heatherington, and R. W. Tabor, unpublished data, 1980). The older of the two ages is consistent with the microfossil stage determination and is probably a better estimate of the absolute age of the Goble Volcanics [see *Armentrout*, 1981].

We sampled the Goble Volcanics on both sides of the younger faults trending south from the Willapa Hills fault zone to help define the timing of the differential rotation shown in the Crescent basement. Eleven sites were collected from roadcuts and quarries along the 40-km length of the Goble outcrop (Figure 9), and tilt corrections were determined from the attitude of local sedimentary interbeds and flow tops. Our experimental procedure was identical to that used for the Crescent Formation described earlier. All sites showed stable single component magnetizations during stepwise AF demagnetization experiments, and Figure 15 is an illustration of typical results. Site mean magnetization directions do not show the systematic domains of rotation displayed by the Crescent sites, although they do show significant between site scatter (Figures 9 and 16 and Table 5). The scatter is probably the

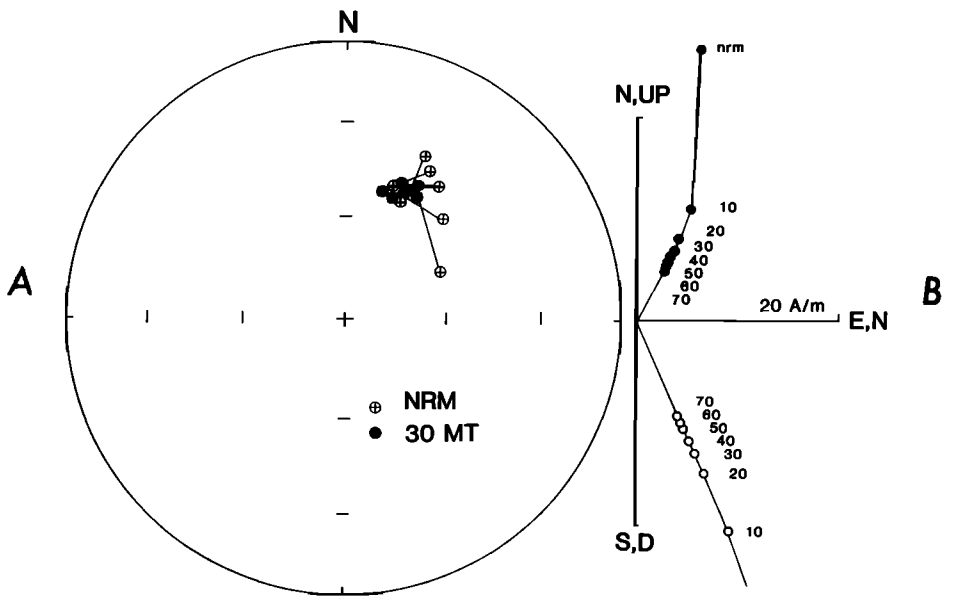



Fig. 15. (*a*) Equal area plot of magnetization direction change during alternating field demagnetization of samples of the Goble Volcanics from flow 0W120. (*b*) Vector demagnetization diagram for sample 0W127, same conventions as in Figure 6.

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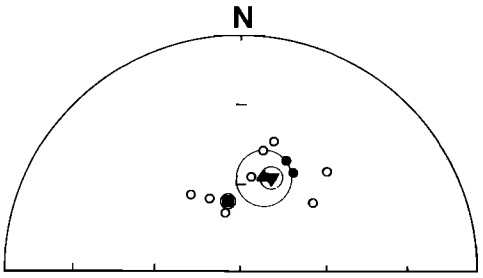



Fig. 16. Equal area plot of site mean magnetization directions for the Goble Volcanics of southwest Washington. Solid circles, normal polarity lower hemisphere; open circles, reversed polarity inverted through origin for comparison. Square is expected Eocene-Oligocene direction from poles of *Diehl et al.* [1983]; upright triangle is mean Goble Volcanics direction from this study; inverted triangle is mean Goble volcanics difection of *Beck and Burr* [1979].

result of local tectonic rotation associated with cross-cutting faults in the volcanics. An extreme example of this may be the northwesternmost site (9W129), which has a southwest declination that could be the result of counterclockwise rotation along the adjacent left-lateral Grays River fault zone. Alternatively, the direction could be the result of an excursion or transition of the magnetic field. In any case, it was not used in the calculation of the formation mean.

The directions pass the fold test at the 90% confidence level and contain both polarities, roughly antipolar. The formation pole is rotated an average of 23° ± 15° east of the North American Eocene and Oligocene reference poles of *Diehl et al.* [1983]. This is similar to the rotation determined for the Goble Volcanics in the foothills of the Cascade Range [*Beck and Burr*, 1979], 50 km to the southeast (Table 6). The shallow inclinations are also similar to the results of *Beck and Burr* [1979] and are not easily explained. Geologic relationships rule out large translation or local tectonostratigraphic problems, but other possible causes of shallow inclination might include magnetic anisotropy related to platy flow structures or demagnetizing effects in thin, flat flow sheets.

The observations that Goble Volcanics lap unconformably across the early boundary structure in the Crescent Formation and do not show the rotation domains found in the Crescent Formation support the conclusion that differential rotations of domains in the Crescent basement occurred during the middle Eocene. The results also show that a significant amount of the total tectonic rotation is of post–late Eocene age and suggest local rotations may be significant. The clockwise rotation for the Goble Volcanics (23° ± 15°, this study; and 28° ± 7°, recalculated from *Beck and Burr* [1979]) are similar to the clockwise rotations for the Crescent Formation in the western Willapa Hills (20° ± 8°) and Black Hills (27° ± 11°, *Globerman et al.* [1982]), and for the western Cascades of Washington (37° ± 9°, *Bates et al.* [1981]). These similarities imply that a regional rotational mechanism of post–late Eocene age has acted on southwest Washington.

## 5. A Model for Deformation and Tectonic Rotation of Southwestern Washington

### *Eocene Tectonics*

In Figure 17 we have tried to depict the sequence of Eocene tectonic events in western Washington. During the early Eocene, oceanic islands and seamounts were rapidly constructed by eruption of tholeiitic and alkalic basalts on oceanic crust. There is evidence that this took place in close proximity to the continental margin. The basalts of the Crescent Formation in the Willapa Hills are interbedded with continentally derived, quartzose, micaceous turbidite sandstones in the upper part of the Crescent section [*Wells*, 1979], and in the Olympic Mountains 100 km to the north, *Cady* [1975] describes continentally derived quartz diorite boulders interbedded in the lower part of the Crescent Formation.

Almost immediately following the eruption of the basalts, or even during the later stages of eruption, the seamounts and islands were sutured to the continental margin (Figure 17*a*). During emplacement the Crescent "plate" was folded and broken into smaller blocks which may have accommodated additional compression by thrusting along the block margins. West-northwest folding and conjugate strike-slip faulting as mapped in the Willapa Hills may have been generated at this time in response to the persistent northeastward convergence of the Farallon plate [*Engebretson*, 1982]. The Willapa Hills were split into an eastern and western block as a result of impingement of the eastern Willapa Hills against the buttress of the Doty Hills block, or as a result of drag against the continental margin during small northward translation of the Willapa Hills (Figure 17*c*). In either case, the eastern Willapa Hills rotated about 32° clockwise with respect to the rest of the Willapa Hills, with the rotation taken up along the EWH fault zone. Also rotated in the EWH were the early formed folds and faults. In the Fort Columbia block (Figure 2), similar clockwise rotations may have occurred as the outboard block got caught in a shear couple between the accreted Coast Range and the rapidly moving Farallon plate. The rotations of the outboard blocks may have been facilitated by the

TABLE 5. Paleomagnetic Data From the Goble Volcanics, Southwest Washington, Tectonically Corrected Site Means

| Site | Demagnetization Field | Site Location | | Remanence Direction | | VGP Position | | Statistics | | | | | Strike and Dip | Rock Type |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Latitude | Longitude | Inclination | Declination | Latitude | Longitude | $N$ | $k$ | $\alpha_{95}$ | ASD | Polarity | | |
| 9W122 | 30 | 46.41 | 236.52 | −55.0 | 227.0 | −53.5 | 145.6 | 7 | 228 | 4.0 | 5.4 | R | 085/24 | sbf |
| 9W051 | 30 | 46.37 | 236.66 | −43.4 | 194.6 | −65.9 | 202.7 | 7 | 228 | 4.0 | 5.4 | R | 166/20 | sbf |
| 9W058 | 30 | 46.37 | 236.66 | −59.7 | 166.7 | −78.9 | 11.4 | 7 | 310 | 3.4 | 4.6 | R | 130/17 | sbf |
| 9W065 | 30 | 46.37 | 236.66 | −58.8 | 147.2 | −65.2 | 322.6 | 7 | 222 | 4.1 | 5.5 | R | 130/17 | sbf |
| 9W136 | 30 | 46.36 | 236.56 | −43.7 | 221.2 | −51.3 | 164.6 | 7 | 166 | 4.7 | 6.3 | R | 080/10 | sbf |
| 0W120 | 30 | 46.34 | 237.02 | 48.4 | 23.4 | 65.1 | 1.7 | 7 | 339 | 3.3 | 4.4 | N | 300/15 | sbf |
| 0W127 | 30 | 46.32 | 237.01 | 50.5 | 29.0 | 63.1 | 350.8 | 7 | 868 | 2.1 | 2.8 | N | 345/12 | sbf |
| 0W106 | 30 | 46.32 | 237.95 | −57.3 | 187.1 | −80.1 | 202.4 | 7 | 94 | 6.2 | 8.4 | R | 120/10 | sbf |
| 0W099 | 30 | 46.30 | 237.98 | −62.6 | 157.4 | −73.9 | 326.8 | 7 | 142 | 5.1 | 6.8 | R | 300/10 | sbf |
| 0W113 | 30 | 46.30 | 236.93 | −46.8 | 190.6 | −69.9 | 208.7 | 7 | 38 | 10.0 | 13.3 | R | 0/0 | sbf |
| 9W129* | 30 | 46.41 | 236.52 | 52.5 | 236.4 | 04.3 | 192.1 | 7 | 150 | 4.9 | 6.6 | N | 100/27 | sbf |

Site descriptions in the work of *Wells* [1982]; sbf, subaerial basalt flow; see Table 1 for explanation of headings.
*Site greater than 2 s.d. from mean not used in calculation of Goble Volcanics mean.

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TABLE 6. Formation Mean of the Goble Volcanics, Southwest Washington

| | Pole Position | | Pole Statistics | | | Direction Statistics | | Remanence Direction | | | | Latitude Shift | Rotation |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Latitude | Longitude | $A_{95}$ | $K$ | $N$ | $\alpha_{95}$ | $k$ | ASD | $\overline{\text{ASD}}$ | Inclination | Declination | $L \pm \Delta L$ | $R \pm \Delta R$ |
| This study, tc | 77.2 | 005.7 | 14.0 | 13 | 10 | 10.3 | 23 | 17.0 | 6.29 | 56.0 | 014.1 | $10.0 \pm 11.2$ | $22.5 \pm 14.5$ |
| This study, unc | 81.4 | 020.0 | 20.1 | 7 | | 14.7 | 12 | 24.1 | | 56.9 | 008.8 | | |
| *Beck and Burr* [1979], tc | 75.5 | 345.5 | 5.5 | | 37 | 4.3 | | | | 57.5 | 018.5 | $7.0 \pm 5.1$ | $28.0 \pm 6.6$ |
| Eocene pole North America | 82.8 | 170.4 | 3.0 | | | | | | | 66.5 | 349.9 | | |
| Oligocene pole, North America | 83.2 | 148.0 | 4.1 | | | | | | | 64.4 | 350.1 | | |

Latitude shift and rotation for the Goble Volcanics are the mean of latitude shift and rotation calculated for both the Eocene and the Oligocene reference poles. Eocene pole covers time period 53–44 m.y.; Oligocene pole covers time period 38–22 m.y. Reference poles from *Diehl et al.* [1983].

underthrusting of accretionary wedge sediments beneath the basalts [*Snavely and Wagner*, 1982].

By middle Eocene time the Coast Range was in place and the late middle Eocene Cowlitz and Skookumchuck (Skookumchuck Formation) deltas had prograded across the tectonic boundary with the continent (Figure 17*d*). Abundant volcanic detritus interbedded with the micaceous arkoses, and tuffs with 40 m.y. potassium-argon ages in the Skookumchuck coal beds [*Triplehorn et al.*, 1980] suggest that the Cascade volcanic arc had already begun to develop. The volcanic Tukwila Formation in the Puget Lowland, dated in part at 42 m.y. [*Frizzell et al.*, 1979] and 44–42 m.y. andesitic dikes in the eastern Olympic Peninsula [*Tabor and Cady*, 1978; P. D. Snavely, unpublished data, 1983] may represent some of the earliest products of the Cascade volcanic arc erupted through the Coast Range basement. This would imply a minimum emplacement age of the Washington Coast Range of about 42 m.y.

### *Post-Eocene Tectonics*

Our Goble paleomagnetic data indicate that a substantial tectonic rotation of the Willapa Hills has occurred since the late Eocene, and other studies show that similar rotation occurred elsewhere in southwest Washington. This suggests the action of a regional tectonic mechanism. The general similarity of post-Eocene rotations in the Oregon and Washington Cascades has been interpreted by some authors as evidence for a rigid microplate consisting of western Oregon and southwest Washington that is rotated clockwise in response to the Miocene opening of the Basin and Range [*Magill and Cox*, 1980; *Magill et al.*, 1981, 1982].

In southwest Washington the striking pattern of post–late Eocene faulting suggests an alternative mechanism to the rotation of a rigid microplate. The pattern is similar to one described by *Freund* [1974] for regions caught in a shear couple along transcurrent fault zones (Figure 18). In Freund's analysis the strain in regions undergoing simple shear is taken up by members of a conjugate secondary shear set (Riedel shears) which rotate in response to the external shear couple. Either one or both of the shear sets ($R$ or $R'$ of Freund) may be active under moderate shear strains; but in an applied dextral shear couple, it is the sinistral $R'$ faults which are nearly orthogonal to the external shear direction that undergo substantial clockwise rotation. The main difficulties with rotation by $R'$ faults are (1) the width of the area undergoing shear increases initially as the fault segments rotate through a position orthogonal to the external shear couple and (2) the amount of fault offset and hence the amount of rotation may be limited as the $R'$ faults rotate into positions normal to the external maximum compressive stress. This mechanism has been used to explain large Neogene clockwise rotations in southern California, the apparent difficulties notwithstanding [*Luyendyk et al.*, 1980]. In southwest Washington we may be able to ignore the difficulties because the width of the region may be relatively unconstrained by the trench on the west, and the amount of post–late Eocene rotation is small ($< 25°$).

In Figure 18 we interpret southwest Washington in terms of a Riedel shear model in which the major N20°W fault zones are the primary expression of simple dextral shear deformation of the fore arc area, while the west-northwest faults are the sinistral $R'$ secondary shears which have accommodated the clockwise rotations. In this model the simple shear would be produced by partial coupling of the fore arc region with the obliquely subducting Farallon plate beneath it.

We can calculate the required dextral displacement and the resultant sinistral displacements on the $R'$ faults from Freund's analysis if we know the amount of rotation, the orientation of the faults, the spacing between the $R'$ faults, and the width of the simple shear domain. Freund (1974) has shown that $\cot f - \cot g = d/w$, where $f$ is the initial angle between $R'$ faults and the external shear direction, $g$ is the same angle after rotation, $d$ is the displacement on the $R'$ faults, and $w$ is the spacing between $R'$ faults (Figure 18). Given a rotation $g - f = 25°$, we find displacement-width ratios ($d/w$) for southwest Washington of 0.49 to 0.72 for presently observed angles ($g$) of 120° to 140°. Thus the resultant sinistral displacements for $R'$ faults should be about 50 to 70% of the spacing between faults, or anywhere from 0.4 to 2.0 km of sinistral displacement for the Fort Columbia–Cathlamet area. A similar analysis can be made for the western Willapa Hills–Doty Hills region using the abundant east-northeast faults as $R'$ faults in place of the apparently rare west-northwest faults. This shows that about 2 km of sinistral offset on east-northeast faults spaced about 4 km apart is required to produce the observed tectonic rotation of about 25° in the western Willapa Hills.

The dextral faults along the boundaries of the rotated regions can be analyzed in a similar manner. The required displacements depend on the width of the rotated area in exactly the same way as do the sinistral faults. To rotate the 100+ km width of the Washington Coast Range along one outboard shear zone would require a minimum of 40 km of dextral slip to produce the observed rotation, assuming no motion occurs without an accompanying increment of rotation. However, the structural picture in southwest Washington suggests several

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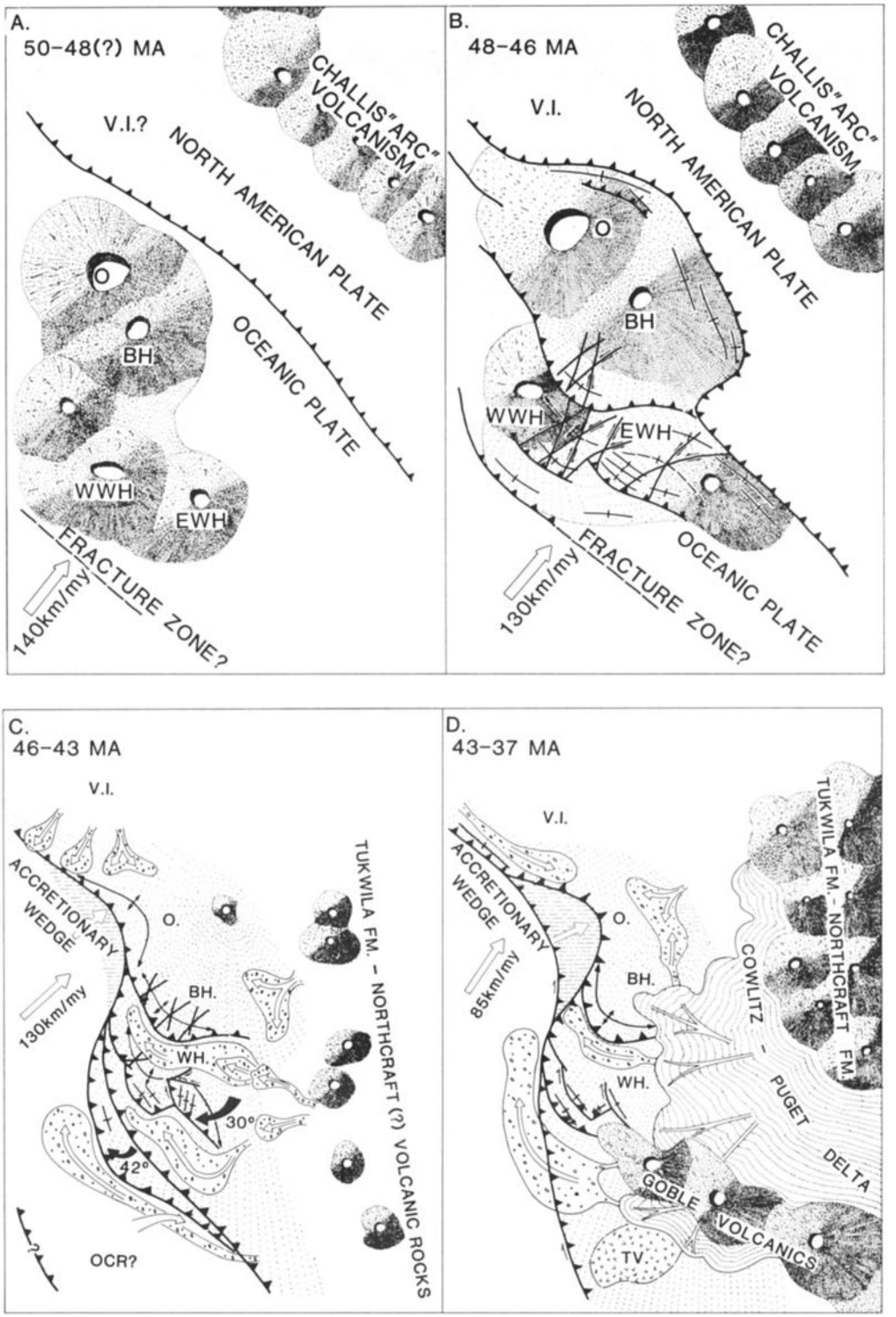



Fig. 17. Model for Eocene tectonic history of southwest Washington: (*a*) 50–48 Ma, growth of basalt shield volcanoes on oceanic crust; V.I., Vancouver Island; O, site of Olympic Mountains; BH, Black Hills site; WWH, Western Willapa Hills site; EWH, Eastern Willapa Hills site; arrow shows direction of Farallon convergence with respect to North America [from *Engebretson*, 1982]; (*b*) 48–46 Ma, initial collision of islands with continental margin; deformation includes folding, thrusting, and strike-slip faulting; (*c*) 46–43 Ma, complete suturing of Coast Range to continental margin; differential rotations in thrust (?) slice and sheared block margins, start of accretionary wedge in Olympic Mountains (?), turbidite sedimentation on Coast Range, initiation of Cascade arc volcanism across old suture; (*d*) 43–37 Ma, continuing shear rotation; prograding Cowlitz-Puget delta (Cowlitz Formation–Puget Group), growth of accretionary wedge, continued folding, eruption of Goble Volcanics and continued Cascade arc volcanism.

dextral fault zones dividing the Coast Range into domains 6 to 20 km in width. The minimum required offset to produce the 25° rotation would then be proportional to the width of the domain.

In Figure 18 we have compared expected displacements from the model with stratigraphic or structural separations observed across the main dextral fault zones and some of the *R'* faults. Most of the faults are poorly exposed, and there are

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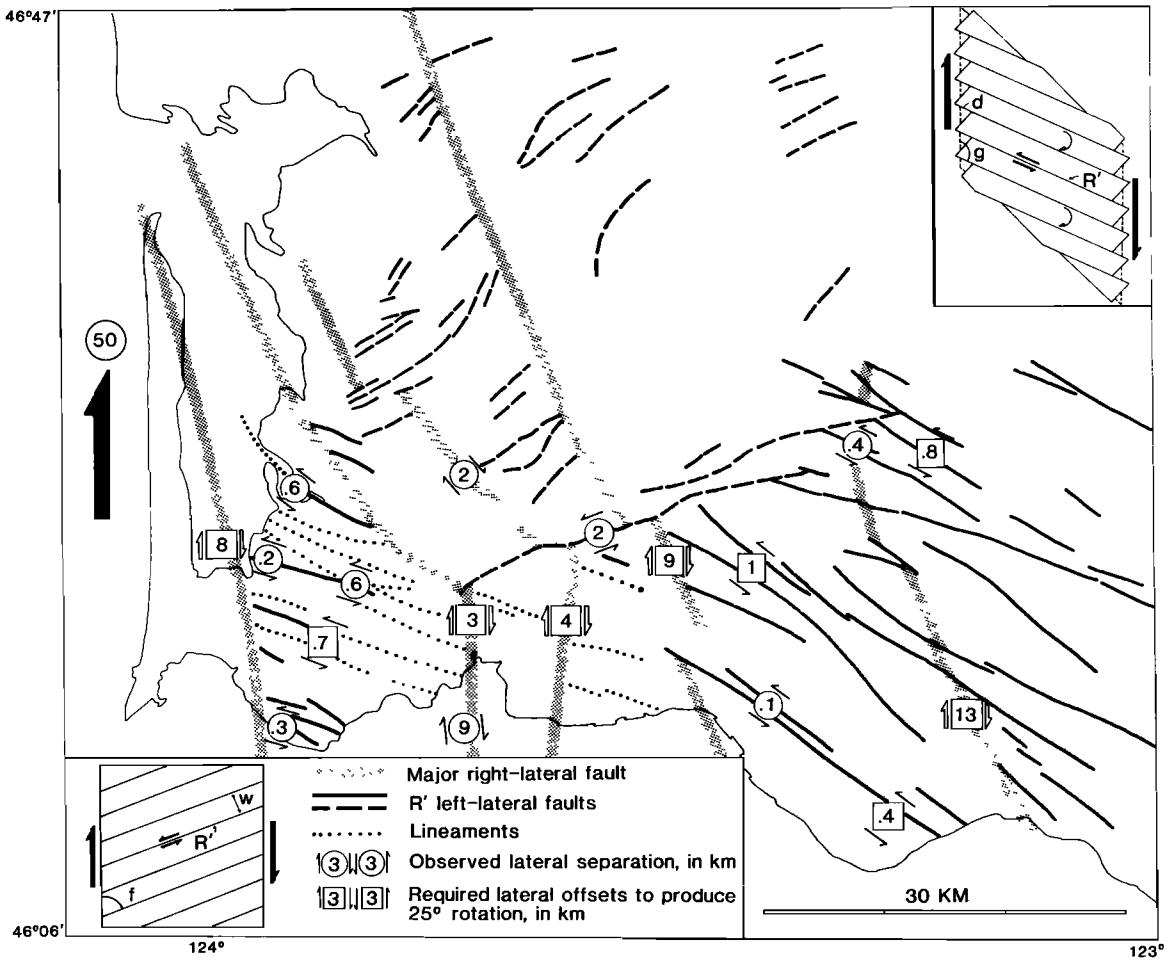



Fig. 18. Strike-slip interpretation of post–late Eocene faulting in southwest Washington and comparison with simple shear deformation model of *Freund* [1974]. Rotation of elongate crustal slices accommodated by sinistral *R'* shears. Large arrow shows northward component of Farallon plate convergence during middle Tertiary time, in kilometers per megaannum, from *Engebretson* [1982]. See text for discussion.

few marker beds that can be correlated across the faults; however, some of the *R'* faults cut distinctive dipping contacts, and maximum sinistral separations can be estimated. Although stratigraphic separation could just as easily be the result of pure vertical movement, the consistent left separation of even steeply dipping contacts and the ubiquitous horizontally grooved fault surfaces suggest a significant lateral component of displacement. The observed offsets on the *R'* faults are about 35 to 100% of the expected offsets in the Fort Columbia area and about 30 to 50% of the expected offsets in the Cathlamet area, although displacements are poorly constrained in the Cathlamet area because the faults are generally striking parallel to the gently dipping strata.

For the dextral slip faults the relatively large predicted displacements seem reasonable for the three western fault zones, although we have meager evidence on which to base our estimates. Large apparent dextral slip may have occurred across the Altoona and Naselle faults, where the middle Miocene basin axis is offset 9 km northward (Figures 3 and 18). Facies relationships and structural trends also change significantly across these faults [*Wolfe and McKee*, 1968, 1972], although their southward continuation across the Columbia River is not obvious.

Further to the east near Cathlamet, large dextral displacements apparently have not occurred since the eruption of the Columbia River Basalt Group (locally of middle Miocene age), although motion may have occurred prior to that time. In order to explain a minimum of 16° clockwise rotation for the Miocene basalts [*Magill et al.*, 1982], about 12 km of dextral slip is required on the Altoona-Naselle fault zone to the west. The observed 9-km dextral separation of the middle Miocene fold axis along the Columbia River is of the right order of magnitude and may have been sufficient to drive the rotations near Cathlamet.

It is also possible that part of the rotation could be generated by a similar mechanism operating on a smaller scale [*Sheriff and Bentley*, 1980; *Sheriff*, 1984]. Closely spaced, small-displacement faults may be very common in southwest Washington, and they would go unmapped beneath the heavy cover of vegetation. Resolution of this possibility awaits detailed paleomagnetic and structural studies of a few well-chosen quarry sites. Although we cannot demonstrate with certainty that Riedel shear rotations have occurred in southwest Washington, there are enough correlations between the observed geology and the model requirements to suggest that in situ post-Eocene shear rotation may be a significant process in this part of the Coast Range.

## 6. Regional Implications

An obvious question is whether local rotation mechanisms might be important elsewhere in the Coast Range. At present, paleomagnetic and geologic data from the rest of the Coast

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Range are not sufficiently detailed to demonstrate the tectonic rotation domains observed in the Crescent Formation of southwest Washington. However, there is some evidence that simple shear rotation accommodated by local faulting has occurred in other Coast Range localities. All tectonically rotated areas for which detailed geologic mapping is available show an abundance of closely spaced west-northwest or east-northeast faults similar to those of southwest Washington (Figure 19). In the unmapped areas, well-developed lineament patterns visible on side-looking radar imagery and topographic maps mimic the mapped faults and suggest similar structural regimes. Although the sense of motion on most of these faults is unknown, mapped stratigraphic separations are consistent with a component of sinistral slip for many of the faults. As in southwest Washington, it seems unlikely that the fault displacements are purely dip-slip because the sense of vertical motion is not consistent across fold axes, and horizontal grooves and slickensides are common on exposed fault surfaces in all of the mapped areas (see references below). These faults may be acting as sinistral $R'$ shears bounding rotating crustal blocks caught in a simple right shear couple.

Detailed geologic mapping by *Snavely et al.* [1976*a*, *b*, *c*] in the central Oregon Coast Range (Figure 19*c*) shows the observed left separation on the west-northwest faults ranges from 0.2 to 4 km with an average of 1.3 km. This suggests that a maximum clockwise rotation of about 20° to 25° could have been accommodated given the mapped fault spacing in this area. In the Tillamook Highlands (Figure 19*b*), compilation of reconnaissance geologic mapping [*Wells et al.*, 1983] shows several large west-northwest faults cross-cutting the Tillamook Volcanics. In some areas these faults are closely spaced, and some have demonstrable sinistral latest motions; however, there is not yet sufficient data to calculate maximum allowable local rotations. Likewise, the lineament patterns in the Black Hills and in the Goble Volcanics of Washington are suggestive but do not permit calculation of a shear rotation component (Figure 19).

Extending significant shear rotations southward as far as the Klamath Mountains is more difficult because of the apparent structural coherence of the middle Eocene Tyee Formation. Paleocurrent indicators in the turbidites show a relatively simple pattern of northward sediment transport from a Klamath Mountains source [*Snavely et al.*, 1964], and published geologic mapping [*Baldwin*, 1974] shows no evidence of structures related to shear rotation. However, Rb-Sr geochemistry of the Tyee Formation suggests an older, more easterly source for the sediments similar to the source terrane of the present Columbia River [*Peterman et al.*, 1981; *Heller*, 1983]. A modest amount of in situ shear rotation (about 25°) in addition to rigid plate motion might allow the paleocurrent directions to reflect a more easterly source without significantly disrupting the observed facies relationships. It would be difficult to detect strike-slip faulting in the Tyee Formation because of the monotonous stratigraphy and gentle folding. However, if Riedel shear rotations have occurred in the southern Oregon Coast Range, then sinistral-slip faults should be found in the Tyee Formation.

More detailed geologic and paleomagnetic studies in the fore arc region of Oregon and Washington are necessary to determine the general applicability of *Freund*'s [1974] shear strain analysis. However, geologic and paleomagnetic investigations in better exposed terrains in southern California along the San Andreas fault [*Luyendyk et al.*, 1980; *Terres et al.*, 1981] and in the eastern Mediterranean along the Dead Sea Rift (H. Ron, personal communication, 1981) suggest that such rotation mechanisms may be ubiquitous in areas undergoing significant shear strain.

*Implications of shear rotation for rigid plate tectonic models.* In situ rotation and nonrigid behavior are likely to be an essential part of the kinematic response of small micro-

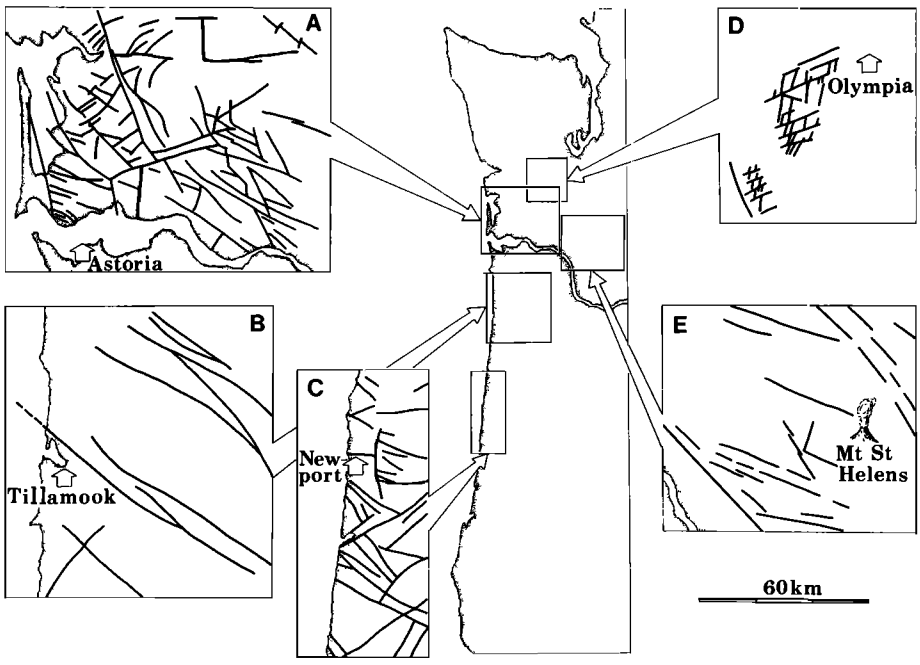



Fig. 19. Fault pattern map for tectonically rotated areas in the Oregon-Washington Coast Range. West-northwest faults are possible $R'$ left-lateral faults, except in Figure 19*d* and in the northern part of Figure 19*a*, where east-northeast faults may be acting as $R'$ shears. (*a*) Southwest Washington, (*b*) Tillamook Highlands, (*c*) central Oregon Coast, (*d*) Black Hills Washington, and (*e*) Goble Volcanics. Geology is from various sources (see text); fault patterns in Black Hills and the Goble Volcanics are lineaments from topographic maps and side-looking radar.

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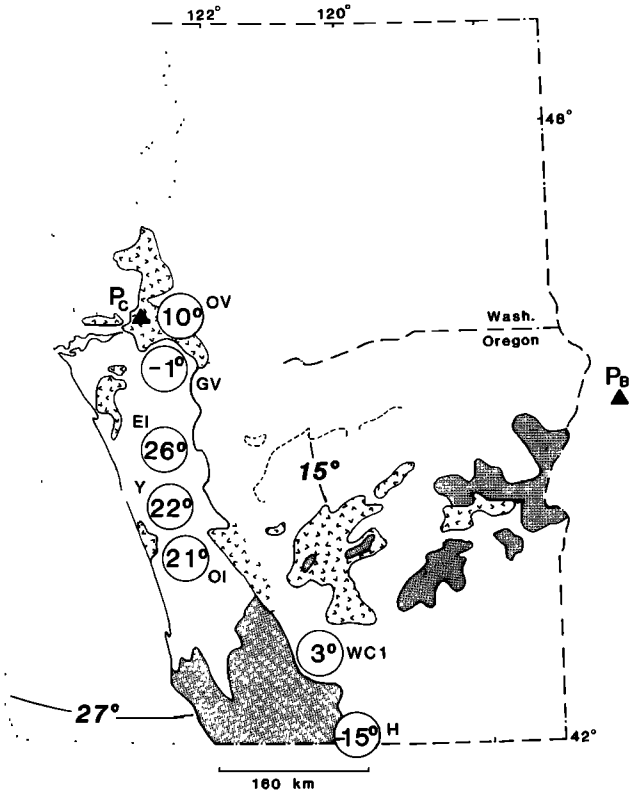



Fig. 20. Rigid plate reconstruction at about 37 Ma limited by small rotation of the Clarno Formation "backstop" on Blue Mountains block; geologic units as in Figure 1, with probable Eocene western Cascade volcanic rocks added. Misfit of paleomagnetic data to model shown in circles; positive numbers indicate additional rotation not accounted for in reconstruction; negative numbers are rotations in excess of observed values. Large deviations from 0° imply small block rotation to make up difference. OV, Volcanic rocks of the Ohanapecosh Formation [*Bates et al.*, 1981]; GV, Goble Volcanics [*Beck and Burr*, 1979]; EI, Eocene intrusive rocks [*Beck and Plumley*, 1980]; Y, Yachats Basalt [*Simpson and Cox*, 1977]; OI, Oligocene intrusive rocks [*Beck and Plumley*, 1980]; WC1, western Cascades [*Magill and Cox*, 1980]; H, Cretaceous Hornbrook Formation [*Mankinen and Irwin*, 1982]. $P_B$ and $P_C$ are pivot points for Blue Mountains and Coast Range, respectively.

plates caught in the boundary zone between two large plates [*Beck*, 1980]. It is reasonable to suggest that the Oregon and Washington fore arc regions have absorbed significant shear strain, given that a large component of convergence parallel to the continental margin has existed for most of Tertiary time [*Engebretson*, 1982]. In contrast to the suggestion of *Magill et al.* [1981] that the lack of surface faulting in western Oregon indicates rigid plate behavior, we would suggest that the lack of faulting in western Oregon and Washington in most compilations is only apparent, a result of the dense cover of vegetation and incomplete coverage of detailed geologic mapping.

Significant local rotation could solve some of the structural and stratigraphic problems associated with models requiring rigid rotating plates. The model of *Magill et al.* [1981] explains the rotation of the Oregon Coast Range as a result of two phases of rigid plate rotation: Phase one involves about 50° of Eocene clockwise rotation around a southern pivot prior to Coast Range accretion; phase two involves about 30° of Miocene and younger clockwise rotation about a northern pivot in response to Basin and Range extension (see *Magill et al.* [1981], Figures 11 and 13, and *Magill and Cox* [1981]). However, the timing of accretion is constrained by the onlap of the continent-derived fluvial-deltaic and shallow marine Cowlitz Formation and correlative units which suture the northern Oregon and southern Washington Coast Ranges to the continent before 40 m.y. ago (*Warren and Norbisrath* [1946], *Snavely et al.* [1958], *Wells et al.* [1983], and our earlier discussion). Because phase one rotation was completed by Cowlitz time, it cannot account for any part of the post-Cowlitz rotations of about 50° indicated by paleomagnetic results from the upper Eocene and Oligocene volcanic and intrusive rocks of the Oregon Coast Range (recalculated from *Simpson and Cox* [1977] and *Beck and Plumley* [1980]). This limits accretion rotation to no more than 20°. It is unlikely that all of the postaccretion rotation of 50° in the Oregon Coast Range was accommodated by phase two plate rotation in response to opening of the Basin and Range. Such a reconstruction would require unrealistic overlaps in the restoration of the Klamath Mountains and Blue Mountains and would also require large Tertiary clockwise rotations in both areas, contrary to newly available paleomagnetic evidence. *Mankinen and Irwin* [1982] and *Craig et al.* [1981] have shown that the Cretaceous Hornbrook Formation in the eastern Klamath Mountains and the overlying Oligocene Cascade volcanic rocks have undergone rather small clockwise rotations of about 15°. Paleomagnetic studies in the middle and upper Eocene and lower Oligocene volcanic rocks of the Clarno Formation in central Oregon also show small clockwise rotations of about 16° [*Beck et al.*, 1978; C. S. Grommé et al., work in preparation, 1984], significantly less than the 50° postaccretion rotations determined for Coast Range rocks of the same age. In a rigid plate model the Clarno provides a backstop for restorative reconstructions of the Coast Range and western Cascade Range (Figure 20). These apparent inconsistencies are most easily explained by nonrigid behavior in the form of local rotation. The subtraction of local rotations from observed Coast Range values would allow less extreme reconstructions and could avoid difficulties with both the Clarno Formation and the Klamath Mountains.

We believe the mechanisms described in this paper contributed a significant component of nonrigid rotation to western Washington and possibly Oregon when viewed at an appropriate regional scale. In fact, on a plate tectonic scale, western Washington and Oregon could be acting as a "soft plate boundary" as envisioned by *Atwater* [1970], rather than a truly rigid plate. Overall, the kinematic behavior of western Oregon and Washington is probably best described as a composite of the rigid plate, collision, and shear rotation models whose elements change with time and place. Initially, the Oregon and Washington Coast Ranges may have been part of separate microplates, with Oregon undergoing up to 20° clockwise rotation prior to complete suturing with North America. Washington was not rotated prior to collision, based on the relationship between early compressional deformation and rotations and on the similarity in the amount of rotation between the oceanic basement rocks and the postaccretion Goble Volcanics. During emplacement of the Coast Ranges, Washington and possibly Oregon were rotated piecemeal by a combination of compressional and shear deformation. Following emplacement, the Coast Ranges and the onlapping early Cascade arc were subject to moderate simple shear rotations in response to continuing oblique subduction. Several episodes of extension in the back arc may have contributed a significant nearly rigid plate rotation component to western Oregon and possibly Washington in postaccretion time; most notable were the middle Eocene block faulting and volcanism in north central Washington [*Davis*, 1980], the implied extension ac-

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companying voluminous Oligocene ash flow tuff eruptions throughout the Basin and Range [*Zoback et al.*, 1981] and the Cascade arc, and finally the post–12 m.y. Basin and Range block faulting [*Zoback et al.*, 1981], which is still occurring today.

## 7. Conclusions

1. Paleomagnetic and structural studies of the middle Eocene submarine basalt basement of the Crescent Formation in the southwest Washington Coast Range show domains of differential rotation ranging in size from a few kilometers across up to at least 30 km across. Rotation of these domains ranges from 20° to perhaps as much as 65° clockwise when compared to the Eocene reference pole for cratonic North America, although the amount of rotation for the smaller domains is poorly constrained by the data.

2. Rotation domains correlate with structural domains bounded by major fault zones which have accommodated the differential rotations. The boundary structures are truncated by unconformably overlying upper middle and upper Eocene sedimentary and volcanic rocks which place the timing of differential rotation of the pillow basalt basement in the middle Eocene.

3. Intense middle Eocene folding in the Crescent is the result of compression of the Coast Range against the North American continental margin. The correlation between increasing clockwise rotation of the paleomagnetic vectors and progressively more easterly orientation of domain fold axes suggests that differential rotation postdates Coast Range suturing and is the result of an early disruption of the newly emplaced marginal terrane. Localized large rotations may represent deformation along the margins of large crustal blocks which are compressed and sheared against one another during accretion.

4. The upper Eocene Goble Volcanics which unconformably overlie the Crescent Formation are rotated about 23° clockwise and do not show the same geographic domains of rotation as the underlying Crescent Formation, although directional scatter is high, suggesting local tectonic rotations are significant. The Goble data when combined with other studies in southwestern Washington suggest regional post-Goble rotations of 25° to 30° have affected most of southwestern Washington.

5. Geologic field investigations show the pattern of post–late Eocene deformation in southwestern Washington to be similar to the deformation envisioned by *Freund* [1974] for areas caught in a simple dextral shear couple along transcurrent faults. Major north-northwest trending faults with several kilometers of dextral slip form the boundaries of cross-faulted shear domains in which the clockwise rotation of elongate crustal slices is accommodated by west-northwest trending sinistral $R'$ Riedel shears. Thirty-five to 100% of the observed post–late Eocene rotations of 25° could have occurred by this shear rotation mechanism.

6. Other rotated areas in the Coast Range and western Cascade Range have a similar fault geometry, and they also may have undergone significant shear rotations. Long-term northward oblique subduction of the Farallon plate beneath the Coast Range throughout most of the Tertiary could have been the driving force for the shear rotations and could explain the rapid eastward decrease in rotation away from the continental margin.

7. Implications of shear rotations for rigid plate tectonic models of the Pacific Northwest are significant. Local Coast Range and Cascade shear rotations could eliminate many of the structural and stratigraphic difficulties associated with large rigid plate rotations. Shear rotation is probably responsible for about half of the total postaccretion rotation, while plate gliding and rotation of western Oregon and Washington off the thermal high of a differentially expanding back arc region (Basin and Range) as suggested by *Magill et al.* [1982] may be responsible for the remainder. The Cenozoic kinematic behavior of western Oregon and Washington is probably best described by a composite model of rigid plate, collision, and shear rotations whose elements change with time and place as the North American plate responds mechanically and thermally to changes in Farallon plate motions and the ever increasing influence of dextral shear along the Pacific plate boundary.

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